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ToE Living Review Letters IV — Part I

**An Introduction to the Mathematical Theory and Core Concepts
of the Theory of Entropicity (ToE): A Rigorous Path Toward a
Complete Derivation of the Einstein Field Equations of General
Relativity as a Limiting Case from an Entropic Field Theory**

*Part I: The Revolutionary Inversion and the Complete Mathematical
Foundations*

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A Rigorous Path Toward a Complete Derivation of the Einstein Field
Equations of General Relativity as a Limiting Case from an Entropic Field
Theory.

Theory of Entropicity (ToE) — Living Review Letters Series.

Letter IV. Part I.

Abstract

This Letter IV of the Theory of Entropicity Living Review Letters Series constitutes the definitive pedagogical and mathematical reference for the Theory of Entropicity (ToE). It consolidates and extends the foundational results of Letters I, II, and III by presenting, for the first time in a single unified treatment, the complete mathematical architecture required to understand ToE from first principles, through to the full derivation of both the left-hand side and right-hand side of the Einstein field equations of general relativity as a limiting case of an entropic field theory. Letter IV is organized into three Parts. Part I, presented here, establishes the revolutionary ontological inversion at the heart of ToE — that entropy is the ontological primitive of physical reality, not matter, energy, or spacetime — and develops in full the mathematical foundations upon which all subsequent constructions rest.

The revolutionary inversion is the central conceptual and logical claim of the Theory of Entropicity. In the standard hierarchy of physics, matter and energy are the primitives, entropy is computed from matter via

Boltzmann's formula $S = k_B \ln W$, and gravity is a geometric field whose source is the matter-energy distribution. ToE inverts this hierarchy: entropy is primitive; matter, energy, radiation, and spacetime are all emergent from the dynamics of the entropic field $S(x)$. This inversion is not a philosophical metaphor but a mathematically precise proposal about the ordering of explanatory and ontological priority. The justification for the inversion is drawn from five independent lines of argument: (i) the Shannon-Landauer-Wheeler information-primacy arguments; (ii) the Bekenstein-Hawking-Jacobson-Verlinde thermodynamic gravity arguments; (iii) the Amari-Čencov information-geometry arguments; (iv) the non-equilibrium thermodynamics arguments; and (v) the logical economy (Occam) argument. These are developed in full in Sections 3 and 4.

Part I then develops the complete mathematical prerequisites needed for ToE. The treatment is deliberately self-contained: beginning with topological spaces and differentiable manifolds (Section 5), the paper progresses to tangent spaces and the tangent bundle (Section 6), cotangent spaces and the cotangent bundle (Section 7), general tensor algebra and the key tensors of ToE including the Fisher-Rao metric, the Obidi Metric, the Ricci tensor, the Einstein tensor, and the Entropic Stress-Energy tensor (Section 8), the complete theory of fiber bundles including vector bundles, principal bundles, connections, and fiber integrals (Section 9), probability distributions as geometric objects on statistical manifolds (Section 10), differential forms and integration on manifolds including Stokes' theorem and the fiber integral formalism for coarse-graining (Section 11), connections and covariant derivatives on curved manifolds including the Levi-Civita connection and the Amari-

Čencov α -connections (Section 12), and the complete curvature theory including the Riemann tensor, the Ricci tensor, the Einstein tensor, the Bianchi identities, and the Obidi Curvature Invariant (OCI) (Section 13).

Special attention is given throughout to moments of probability distributions (Section 14), which form the bridge between the microscopic information-geometric picture and the macroscopic tensorial structure of general relativity. In particular, the identification of the second moment of the entropic momentum distribution with the Entropic Stress-Energy Tensor $T_{\mu\nu}^{ent}$ is elaborated in full, and the fiber integral interpretation of kinetic theory coarse-graining is made mathematically precise.

All mathematical objects introduced in Part I are defined rigorously, illustrated with concrete physical examples drawn from both classical physics and the ToE framework, and accompanied by Scholia — formal remarks in the tradition of Newton's Principia and the prior Letters of the ToE LRLS — that articulate the physical and philosophical meaning of each result. The paper closes with a complete summary, a comprehensive table of mathematical symbols and notation, and a full reference list covering all primary sources in information geometry, thermodynamic gravity, and the prior ToE Letters.

Keywords: Theory of Entropicity (ToE); Obidi Action; Obidi Transformation; Obidi Metric; Obidi Curvature Invariant; Entropic Field; Entropy as Ontological Substrate; Statistical Manifolds; Fisher-Rao Metric; Fubini-Study Metric; Amari-Čencov α -connections; Kullback-Leibler Divergence; Araki-Umegaki Divergence; Rényi-Tsallis Entropy; Fiber Bundles; Tangent Bundle; Cotangent Bundle; Tensor Fields; Probability Distributions; Fiber Integrals; Moments of Distributions; Hybrid Metric-Affine Space; Master Entropic Equation; Vuli-Ndlela Integral; General Relativity; Einstein Field Equations; Emergent Gravity; Revolutionary Inversion



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Section 1. Preface — Purpose and Scope of Letter IV

This is Letter IV of the Theory of Entropicity Living Review Letters Series. It is, by design, the longest, most comprehensive, and most pedagogically thorough Letter yet produced within this series. The author wishes to be clear from the outset about why this is so, and about what this Letter is trying to accomplish, because the goals of Letter IV are somewhat different from those of Letters I, II, and III, and that difference determines everything about how this Letter is organized and written.

Letters I, II, and III established the Theory of Entropicity. They introduced the fundamental claim — that entropy is the ontological primitive of physical reality — derived the core objects of the theory (the Obidi Action, the Obidi Transformation, the Obidi Metric, the Obidi Curvature Invariant, the Hybrid Metric-Affine Space, the Master Entropic Equation, and the Vuli-Ndlela Integral), and sketched, in varying degrees of completeness, the path from the entropic field to the Einstein field equations of general relativity. Those Letters were addressed to the frontier: to experts who already know the mathematics and want to see the new results.

Letter IV is addressed to a different set of goals. Its first purpose is completeness: to place, for the first time in a single unified treatment, all of the mathematics that ToE requires. Not a sketch. Not a reference list. Not an invitation to consult standard textbooks. The full treatment, here, in this

document. A reader who works through Letter IV, Section by Section, should arrive at the end equipped with every mathematical tool needed to read, verify, and extend any result in the prior Letters and in Parts II and III of Letter IV itself.

There are three audiences that this Letter specifically addresses, and it is important to name them because their needs are different and the author has tried to serve all three simultaneously:

The first audience consists of advanced students — doctoral students in theoretical physics, mathematics, or mathematical physics — who are encountering the Theory of Entropicity for the first time and want to understand it properly. These readers know calculus and linear algebra. They have likely seen some differential geometry. But they have not worked through information geometry, and they may not have seen the precise mathematical relationship between probability distributions, the Fisher-Rao metric, and emergent Riemannian structure. For these readers, Letter IV is a guided mathematical journey from familiar territory (smooth manifolds, tensor fields) into the newer terrain of information geometry and entropic field theory. Every concept is introduced from first principles. Every definition is followed by at least one concrete physical example. Every major theorem is accompanied by a Scholium that draws out its physical meaning.

The second audience consists of researchers in theoretical physics — people who work on quantum gravity, string theory, loop quantum gravity, or related approaches — who want to assess the logical foundations of ToE critically and carefully. These readers are experts. They know general relativity thoroughly. They know quantum field theory. They may or may not know information geometry. What they want from this Letter is a completely explicit, logically transparent account of every step in the ToE derivation:

which assumptions are made, where they are made, what they imply, and how they lead to the Einstein field equations. Letter IV is written to satisfy these readers by making every step explicit, by clearly distinguishing rigorous results from conjectures, and by locating the precise point where ToE's assumptions differ from those of other approaches to quantum gravity.

The third audience consists of mathematicians and information theorists — people who work on statistical manifolds, information geometry, optimal transport, or non-equilibrium thermodynamics — who want to understand how the information-geometric structures they know and work with connect, in ToE, to the geometry of physical spacetime and the source of gravitational curvature. For these readers, Letter IV provides the physical interpretation of every information-geometric object: what the Fisher-Rao metric means in terms of forces and curvature, why the α -connections correspond to irreversible dynamics, and precisely how the fiber integral over momentum space produces the stress-energy tensor that sources Einstein's equations.

No prior knowledge of the Theory of Entropicity is assumed. A reader who has not read Letters I through III can begin with Letter IV and obtain a complete, self-contained introduction to the theory. The prior Letters are referenced throughout, and their key results are stated and used, but their derivations are not a prerequisite for following the present text. What is assumed is basic mathematical maturity: familiarity with calculus in several variables, elementary linear algebra, and the rudiments of classical mechanics and electrodynamics at the level of an advanced undergraduate or early doctoral student.

It is important to understand what the Theory of Entropicity is claiming, and what it is not claiming, because the claim is sometimes misread in two opposite directions. Some readers, encountering the phrase "entropy is the

ontological primitive," imagine that ToE is proposing a form of idealism or anti-realism — that the physical world is somehow "made of information" in a metaphorical or subjective sense. This is not what ToE claims. The entropic field $S(x)$ is as objective and as physical as the electromagnetic field $F_{\mu\nu}$. It has definite values at each point of the manifold. It satisfies a definite equation of motion. It has measurable consequences for the geometry of spacetime and the motion of matter. There is nothing subjective or metaphorical about it.

Other readers, encountering the mathematical structure of the Fisher-Rao metric and the information-geometric apparatus, imagine that ToE is proposing that physics is merely statistics — that the laws of nature are really just statements about our ignorance of underlying classical configurations. This too is not what ToE claims. The entropic field is not a measure of ignorance. It is an objective physical field. Its geometric properties are real geometric properties of an objective structure. The fact that this structure can be described using the mathematics of probability theory does not make it subjective any more than the fact that quantum mechanics uses probability amplitudes makes quantum mechanics merely a theory about our knowledge of particles.

ToE's most important claim is this: the ordering of explanation must be reversed. Instead of asking "given matter, what is its entropy?", we must ask "given the entropy field, what is matter?" This is a precise mathematical statement about which objects are axioms and which are theorems in the fundamental theory of nature. Establishing that reversal, and carrying it through rigorously to the recovery of general relativity as a limiting case, is the central project of this Letter IV and of the Letter series as a whole.

The present Part I covers the revolutionary inversion and the mathematical foundations. Part II (to follow) will cover the complete information geometry

(Fisher-Rao, Fubini-Study, α -connections, divergences) and the complete ToE core structure (Obidi Action, Obidi Transformation, Obidi Metric, OCI, HMAS, MEE, Vuli-Ndlela Integral). Part III will present the complete derivation of the Einstein field equations as a limiting case and provide the comprehensive literature review situating ToE within the broader landscape of quantum gravity approaches.

A note on notation: the Einstein summation convention is used throughout — repeated upper and lower indices are summed over. Greek indices μ, ν, ρ, σ run from 0 to 3 and label spacetime coordinates. Latin indices i, j, k run from 1 to 3 and label spatial coordinates. Capital Latin indices A, B, C label coordinates on the information manifold or statistical manifold. The signature convention for the Lorentzian metric is $(-, +, +, +)$. Natural units with $c = 1$ are used unless otherwise specified. A complete table of symbols is given in Section 16.



Section 2. Introduction — The Need for a New Foundation

The twentieth century bequeathed to physics two magnificent and apparently complete theories of nature. General relativity (GR), developed by Einstein between 1905 and 1915 and refined through a century of experimental testing, describes the geometry of spacetime and the dynamics of gravitational fields. Quantum mechanics (QM), developed between 1900 and 1930 by Planck, Bohr, Heisenberg, Schrödinger, Born, Dirac, and many others, describes the behavior of matter and radiation at microscopic scales. Both theories have been tested to extraordinary precision. GR correctly predicts the precession of Mercury's perihelion, the deflection of starlight by

the Sun, gravitational redshift, gravitational wave emission from binary systems, and the large-scale structure of the cosmos. QM, extended to the relativistic quantum field theories of the Standard Model, correctly predicts the anomalous magnetic moment of the electron to eleven significant figures, the masses and decay rates of hundreds of particles, and the entire periodic table of elements. By any measure of scientific success, these theories are triumphs.

And yet, at the deepest level, general relativity and quantum mechanics are not merely incomplete — they are fundamentally incompatible. The incompatibility is not a technical difficulty that has resisted resolution due to insufficient cleverness. It is a structural clash between the foundational assumptions of the two theories, a clash that becomes catastrophic precisely at the scale — the Planck scale — where both theories are simultaneously relevant.

General relativity treats spacetime as a smooth, classical, four-dimensional pseudo-Riemannian manifold $(M, g_{\mu\nu})$. The metric $g_{\mu\nu}$ encodes both the geometry of spacetime and the gravitational field. It is a smooth, deterministic, classical field that satisfies Einstein's field equations:

(2.1)

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$$

$(1/2) g$

$\mu\nu$

$R = (8$

π

G/c

4

$) T$

$\mu\nu$

Here $R_{\mu\nu}$ is the Ricci curvature tensor, $R = g^{\mu\nu}R_{\mu\nu}$ is the Ricci scalar, $G_{\mu\nu}$ is the Einstein tensor, G is Newton's gravitational constant, and $T_{\mu\nu}$ is the stress-energy tensor of matter and radiation. This equation says: the geometry of spacetime (left side) is determined by the distribution of matter and energy (right side). The metric $g_{\mu\nu}$ is the fundamental field of GR. It is treated as a definite, precise, classical quantity.

Quantum mechanics, on the other hand, describes matter as fundamentally probabilistic. A particle does not have a definite position or momentum; it has a wavefunction $\psi(x, t)$, which is a complex-valued function whose modulus squared $|\psi(x, t)|^2$ gives the probability density of finding the particle at position x at time t . Observable quantities are represented by Hermitian operators acting on a Hilbert space, and measurements yield eigenvalues of these operators with probabilities given by Born's rule. The theory is fundamentally non-commutative: position and momentum operators satisfy $[\hat{x}, \hat{p}] = i\hbar$, with the consequence that they cannot both be simultaneously specified with arbitrary precision (Heisenberg's uncertainty principle).

The fundamental problem of quantum gravity is this: if matter is quantum-mechanical, and if the Einstein field equations say that spacetime geometry is sourced by matter, then what happens to spacetime geometry when matter is in a superposition? The right-hand side of equation (2.1) — the stress-energy tensor $T_{\mu\nu}$ — becomes a quantum operator. But the left-hand side — the Einstein tensor $G_{\mu\nu}$ — is a classical geometric quantity. The equation then says: a classical geometric object equals a quantum operator. This is not merely uncomfortable. It is, in a precise mathematical sense, incoherent. One cannot equate a definite real number with a quantum observable that takes different values in different branches of a superposition.

Attempts to resolve this incompatibility have occupied theoretical physics for nearly a century. String theory proposes that the fundamental constituents of nature are one-dimensional strings whose vibrational modes produce all known particles and forces, including a spin-2 graviton that is the quantum of the gravitational field. Loop quantum gravity proposes that spacetime itself is quantized at the Planck scale, built up from discrete spin-network structures. Causal dynamical triangulations, causal sets, non-commutative geometry, and dozens of other approaches have been proposed and investigated at length. All of these approaches share a common feature: they accept the standard hierarchy of physics.

What is the standard hierarchy? In physics as currently practiced, the fundamental objects are:

- **Matter and radiation:** particles with mass, charge, and spin; quantum fields carrying energy and momentum.
- **Spacetime:** the manifold on which matter lives, whose geometry is determined by the Einstein field equations.

- **Fundamental constants:** G , $\hbar$, c , k_B — empirical parameters that must be measured but cannot be derived from more fundamental principles.
- **Entropy:** a derived quantity, computed from the number of accessible microstates of a material system via Boltzmann's formula $S = k_B \ln W$.
- **Information:** a descriptor of our knowledge of the arrangement of matter.

In this hierarchy, entropy is downstream of matter. You first have matter; matter has microstates; entropy is a measure of how many microstates are consistent with the macroscopic constraints you observe. Information is even further downstream: it describes what you happen to know about which microstate the matter is actually in.

The Theory of Entropicity proposes that this hierarchy is upside down, and that the incompatibility between general relativity and quantum mechanics is a symptom of the inversion. The argument is not primarily philosophical. It is physical and mathematical. A series of results in statistical mechanics, thermodynamics, and information theory over the past eighty years have systematically revealed that entropy and information are not derived quantities but foundational ones — that the deep structure of physical law is written in the language of information geometry, and that what we call "matter," "energy," and "spacetime" are macroscopic emergent structures arising from the underlying entropic dynamics.

The key results pointing toward this conclusion are the following. Shannon's 1948 derivation of information entropy showed that there is a unique measure of the information content of a probability distribution, and that this measure satisfies the same formal properties as Boltzmann-Gibbs entropy in statistical mechanics — a deep structural link between information and

thermodynamics. Landauer's 1961 principle established that erasing one bit of information necessarily generates at least $k_B T \ln 2$ of heat — making information thermodynamically concrete and costly. Wheeler's "It from Bit" conjecture proposed that every particle of matter and every force of nature derives its existence from binary observational responses to yes-or-no questions. These three results collectively establish the information-primacy thesis: that information is not merely a description of matter but has a prior claim on physical reality.

From the direction of gravity, the argument is equally compelling. Bekenstein's 1973 discovery that black holes carry entropy proportional to their event horizon area — $S_{BH} = k_B A / (4\ell_P^2)$ — was revolutionary. Entropy, previously thought to be a property of matter in some bounded region of space, turned out to be a property of the boundary of a spacetime region, independent of what matter is inside. Hawking's 1974-1975 calculation that black holes radiate thermally at temperature $T_H = \hbar c^3 / (8\pi G M k_B)$ deepened this: the gravitational field itself — without any obvious matter content — has thermodynamic properties. The gravitational field has a temperature. Jacobson's 1995 result was the most direct: he showed that the Einstein field equations themselves — the full nonlinear equations of general relativity, not an approximation — follow from the thermodynamic relation $\delta Q = T dS$ applied to a local Rindler horizon, combined with the Bekenstein entropy-area relation. In other words, Einstein's equations are not a law about geometry; they are a thermodynamic equation of state. Verlinde in 2011 extended this by showing that Newton's gravitational force law can be derived as an entropic force from the gradient of entropy on holographic screens.

These results do not merely suggest that gravity is consistent with thermodynamics. They show that gravity IS thermodynamics, in a deep and precise mathematical sense. If that is so, then the question becomes: which

thermodynamics? What is the entropic variable that sources spacetime geometry? What is the field equation that governs it? These are the questions that the Theory of Entropicity answers.

ToE's answer is to elevate the local entropy density $S(x)$ to a primary dynamical field with its own equation of motion — the Master Entropic Equation (MEE) — and to derive spacetime, matter, and the Einstein field equations from the dynamics of this field. The MEE is derived from the Obidi Action by the principle of extremal entropy. The Obidi Metric, which gives spacetime its geometry, is derived from the Fisher-Rao information metric of the underlying statistical manifold by the Obidi Transformation — a specific algebraic procedure that converts a positive-definite information-geometric metric into a Lorentzian spacetime metric. The Einstein field equations emerge in the limit of small entropy gradients and near-equilibrium configurations.

This Letter IV exists to make every step in this program mathematically precise, pedagogically accessible, and logically transparent. Part I begins at the beginning: with the revolutionary inversion and with the mathematical language required to state and prove the theory's claims.



PART I: THE REVOLUTIONARY INVERSION

Section 3. The Standard Hierarchy and Why It Must Be Inverted

To understand what the Theory of Entropicity proposes, one must first understand precisely what it is inverting. The standard hierarchy of modern physics is not a single explicit postulate but an implicit ordering of explanatory and ontological priority that is embedded in the structure of every major physical theory. To articulate it clearly is already to see its vulnerability.

3.1 The Standard Hierarchy Stated

In the standard picture, physical reality is organized as follows. At the most fundamental level are the quantum fields of the Standard Model: the quark fields, the lepton fields, the gauge boson fields (photon, W, Z, gluons), and the Higgs field. These fields exist on a background spacetime manifold and interact according to Lagrangian dynamics specified by the Standard Model. Their properties — masses, charges, spins, coupling constants — are given by fiat: they are the parameters of the theory, measured by experiment, not derived from any deeper structure.

Gravity is handled separately. The gravitational field is the metric tensor $g_{\mu\nu}$, which satisfies the Einstein field equations sourced by the total stress-energy of all matter and radiation. In the classical limit, this is:

(3.1)

G
 $\mu\nu$

$$+ \Lambda \int d^4x \sqrt{-g} g^{\mu\nu} = (8\pi G/c^4) T^{\mu\nu}$$

where Λ is the cosmological constant. Entropy enters the picture only after matter has been specified: given the microscopic constituents of a system, entropy S is computed from the number of accessible microstates W by Boltzmann's formula:

(3.2)

$$S = k_B \ln W$$

Information is yet further derived: it describes how much we know about which of the W microstates the system is actually in. The sequence of

derivation runs: fundamental quantum fields $\rightarrow$ microscopic states $\rightarrow$ macroscopic observables $\rightarrow$ entropy $\rightarrow$ information. Entropy and information are near the end of this chain, not the beginning.

3.2 Five Arguments for the Inversion

3.2(a) The Entropy-Information Primacy Argument

Claude Shannon, in his landmark 1948 paper "A Mathematical Theory of Communication," proved that there is essentially a unique measure of the information content of a probability distribution over a finite set of outcomes, given three natural requirements: continuity in the probabilities, symmetry (relabeling outcomes does not change the amount of information), and additivity (independent experiments contribute additive amounts of information). That unique measure is:

(3.3)

$$H(p) = - \sum_i p_i \log p_i$$

where the sum runs over all outcomes i and p_i is the probability of outcome i . This is formally identical to the Boltzmann-Gibbs entropy of statistical mechanics (with k_B set to unity and the logarithm base chosen appropriately). This formal identity is not a coincidence or a mere analogy. It reveals a deep structural unity between information theory and thermodynamics: the entropy of a thermodynamic system is the information entropy of the probability distribution over its microstates.

Rolf Landauer, in his 1961 paper "Irreversibility and Heat Generation in the Computing Process," proved that the erasure of one bit of information is a physically irreversible process that necessarily dissipates at least $k_B T \ln 2$ of energy as heat into the environment. This result — known as Landauer's Principle — establishes that information is not merely an abstract mathematical object. Information has a definite thermodynamic cost. Erasing it costs energy. This means that information is physical: it is as real, as objective, and as consequential as energy or momentum.

John Archibald Wheeler, in his 1990 essay "It from Bit," articulated the conjecture that every particle of matter, every field of force, even the spacetime continuum itself, derives its existence and meaning from answers to binary yes-or-no questions — bits of information. Every physical thing, in Wheeler's vision, is at bottom an information-theoretic entity. This is not merely a philosophical position. It is the natural extrapolation of the Bekenstein-Hawking result: if the information content of a spatial region is bounded by its boundary area (the holographic principle), then the fundamental constituents of that region are not volume degrees of freedom (particles, fields in the interior) but boundary degrees of freedom (bits of information on a holographic screen).

These three results collectively establish the information-primacy thesis: information is not derived from matter. Information is prior to matter, both logically and physically. If so, then the entropy function S , which measures information, is prior to the matter distributions from which it is ordinarily computed. The standard hierarchy has it backward.

3.2(b) The Thermodynamic Gravity Arguments

The thermodynamic structure of gravity was revealed in a series of foundational results spanning from 1973 to 2011. Jacob Bekenstein showed in 1973 that a black hole of horizon area A carries entropy:

(3.4)

$$S_{\text{BH}} = k_B \frac{A}{4 \ell_P^2}$$

where $\ell_P = (\hbar G/c^3)^{1/2}$ is the Planck length. This result is remarkable for two reasons. First, the entropy is proportional to the area of the horizon, not the volume of the black hole interior. This violates the usual extensivity of entropy in thermodynamics (entropy is normally proportional to volume). Second, the

formula shows that the gravitational field has a definite, objective entropy that depends on its geometric properties (the horizon area), not on any particular distribution of matter.

Stephen Hawking showed in 1974–1975 that black holes emit thermal radiation at temperature:

(3.5)

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}$$

This result, derived using quantum field theory in curved spacetime, shows that the gravitational field — embodied in the geometry of the black hole's exterior spacetime — has a definite temperature. Not the matter that formed the black hole, which has long since been subsumed — the *gravitational geometry itself* has a temperature. This is extraordinary: temperature is a

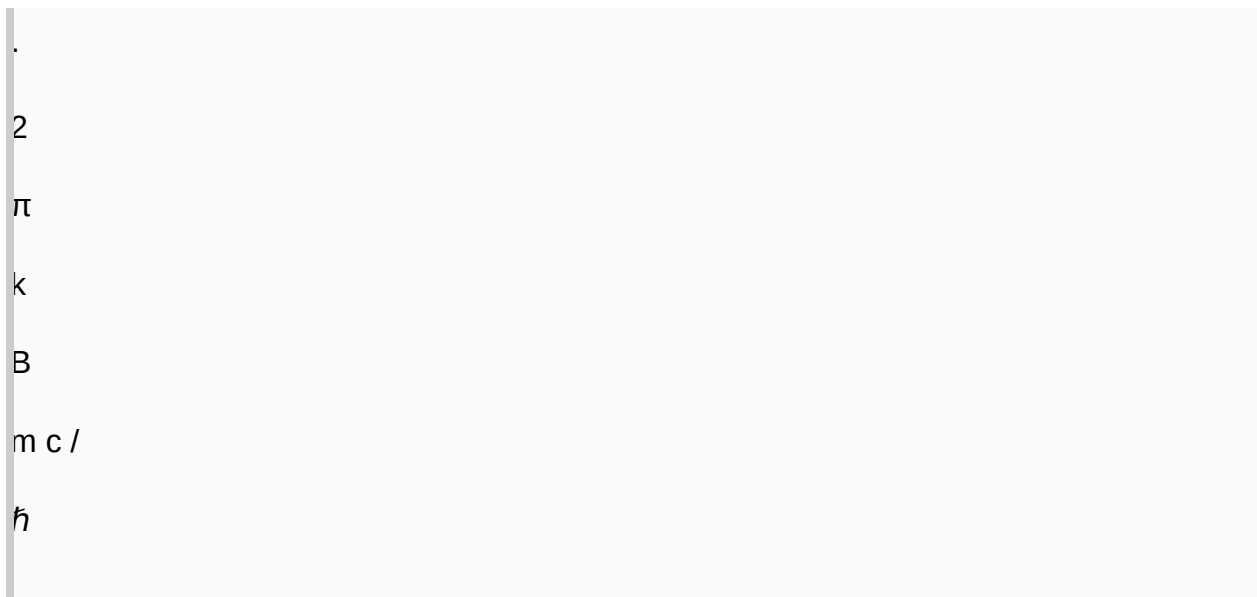
thermodynamic property, and here it belongs to a geometric object. The gravitational field is thermodynamic.

Ted Jacobson's 1995 result is the most profound of all. He showed that the Einstein field equations (3.1), in their full nonlinear generality, follow as a thermodynamic equation of state from: (i) the Bekenstein entropy-area relation (3.4) applied to local Rindler horizons; (ii) the local thermodynamic relation $\delta Q = T dS$ where δQ is the heat flux through the horizon and T is the Unruh temperature seen by a locally accelerating observer; and (iii) the assumption that the entropy is proportional to the area for all local causal horizons. The Einstein field equations are not a law of geometry. They are an equation of state for the geometry of spacetime, in the same sense that the ideal gas law $PV = nRT$ is an equation of state for a gas: it encodes the thermodynamic equilibrium condition of the underlying microscopic degrees of freedom.

Erik Verlinde showed in 2011 that Newton's second law and the law of gravity follow from the entropic force on a holographic screen. If a particle of mass m approaches a holographic screen by a displacement Δx , and if the entropy on the screen changes by $\Delta S = 2\pi k_B m c \Delta x / \hbar$, then the entropic force is:

(3.6)

$$F = T \left(\frac{\Delta S}{\Delta x} \right) = T$$



which, combined with the Unruh temperature $T = \hbar a / (2\pi k_B c)$, gives $F = ma$, and from the equipartition of energy on the holographic screen, gives Newton's law of gravitation. Gravity is an entropic force: it arises because systems move in the direction of increasing entropy. This is not an approximation; it is exact in the appropriate framework.

Together, these four results constitute a coherent and powerful case: gravity is not a fundamental force but a thermodynamic phenomenon arising from an underlying entropic structure. The Einstein field equations are thermodynamic equations of state. This is the thermodynamic gravity argument for the revolutionary inversion.

3.2(c) The Information-Geometry Argument

A family of probability distributions $\{p(x/\theta)\}$ labeled by parameters $\theta = (\theta^1, \dots, \theta^n)$ forms a statistical manifold. Shun-ichi Amari and Hiroshi Nagaoka showed, building on earlier work by Calyampudi Radhakrishna Rao (1945) and Ronald Fisher (1925), that this statistical manifold carries a natural Riemannian metric — the Fisher-Rao metric — given by:

(3.7)

g

F

i, j

$($

θ

$) =$

Σ

x

$p(x|$

θ

$) ($

∂

i

$\log p(x|$

θ

$)) ($

∂

i

$\log p(x|$

θ

$))$

This metric measures the distinguishability of nearby distributions: two distributions are "far apart" in the Fisher-Rao geometry when small changes in parameters produce large, statistically distinguishable differences in the distribution. The metric is positive definite: it is a genuine Riemannian metric on the space of probability distributions.

Nikolai Čencov proved in 1982 that the Fisher-Rao metric is essentially the unique metric on statistical manifolds that is invariant under Markov (congruent) embeddings — meaning that it is the unique metric consistent with the fundamental operations of statistical inference. The Fisher-Rao metric is not an arbitrary choice imposed on the statistical manifold; it is the canonical, invariantly defined geometry of the space of probability distributions.

The information-geometry argument for the revolutionary inversion is this: if physical states are described by probability distributions (which quantum mechanics says they are), then those states automatically carry natural geometric structure — the Fisher-Rao geometry. The question is not whether information produces geometry; by Čencov's theorem, it does, inevitably and canonically. The question is whether that geometry can be made physical — whether it can be identified with, or used to derive, the physical geometry of spacetime. ToE answers: yes, via the Obidi Transformation.

3.2(d) The Non-Equilibrium Thermodynamics Argument

In non-equilibrium thermodynamics, the local entropy density $s(x, t)$ (entropy per unit volume) is already a recognized physical field. The second law of thermodynamics, stated in local form, is:

(3.8)

$$\frac{\partial}{\partial t} s + \nabla \cdot J_s = \sigma_s \geq 0$$

where J_s is the entropy current density and σ_s is the local entropy production rate. This is a field equation for the entropy density field $s(x,t)$. It is already being treated as a dynamical field in non-equilibrium thermodynamics. What ToE does is simply to take this seriously and extend it: instead of treating (3.8) as a derived consequence of underlying matter dynamics, ToE promotes $S(x)$ to a primary dynamical field with its own fundamental equation of motion, from which the matter dynamics is derived.

This is not a radical step in form. It is analogous to what Maxwell did with the electromagnetic field: instead of treating the electromagnetic field as a derived effect of charged particles, Maxwell promoted it to an independent dynamical entity with its own field equations. The conceptual move is the same; only the field is different. ToE does for the entropy field what Maxwell

did for the electromagnetic field: gives it independent dynamical status and derives the behavior of matter from its dynamics.

3.2(e) The Logical Economy Argument

The standard hierarchy of physics requires, as irreducible primitives: the quantum fields of the Standard Model (at least seventeen fundamental fields), the spacetime manifold, the gravitational field, the cosmological constant, approximately twenty-five independent coupling constants and particle masses, and three fundamental constants (G , $\hbar$, c) whose values are given by measurement but not explained by any more fundamental principle. This is an enormous number of unexplained primitives.

The ToE hierarchy requires, as its single irreducible primitive, the entropic field $S(x)$ on a statistical manifold, governed by the Obidi Action. From this single primitive, ToE aims to derive: the spacetime metric (as the Obidi Metric); the matter content (from the localized, stabilized entropic excitations); the coupling constants (as ratios determined by the entropy of the underlying binary choices, characterized by the $\text{OCI} = \ln 2$); the cosmological constant (from the vacuum entropy density); and the Einstein field equations (as the near-equilibrium limit of the Master Entropic Equation). Whether ToE fully achieves this program in all details is a question for Parts II and III; but the aspiration is not unreasonable, and the logical economy argument strongly motivates exploring it. If a single primitive can replace twenty-five, the hypothesis deserves the most serious investigation.

Scholium 3.1 — On the Physical Meaning of the Revolutionary Inversion

The revolutionary inversion is the claim that entropy is ontologically prior to matter — that $S(x)$ is the primary object and that matter, energy,

spacetime, and all other physical quantities are derived from it. It is important to understand precisely what this claim does and does not entail, because it is frequently misread in either of two opposite directions, both of which miss the point.

The inversion does NOT mean that physics reduces to abstract information theory in a manner devoid of physical content. The entropic field $S(x)$ is as objective, as measurable, and as physically real as the electromagnetic field $F_{\mu\nu}$ or the gravitational field $g_{\mu\nu}$. It has definite values at each point of the manifold. It satisfies a definite equation of motion. It carries energy and momentum (via the Entropic Stress-Energy Tensor). It curves spacetime. These are not metaphorical statements. They are the concrete physical content of the Master Entropic Equation and the Obidi Action. The word "entropic" in "entropic field" is not a qualification that makes the field less real; it is a specification of the type of field under consideration, just as "electromagnetic" is a specification of the type of the field $F_{\mu\nu}$.

The inversion does NOT mean that physics is "really" just statistics in the sense of a theory of our ignorance of classical microstates. This reading confuses the mathematical framework with the physical content. Yes, ToE uses the mathematics of probability theory. But so does quantum mechanics. The fact that quantum mechanics uses probability amplitudes does not make it a theory of ignorance about hidden classical particles; it makes it a theory about genuinely probabilistic quantum states. Similarly, ToE uses information geometry not because physical reality is a matter of our knowledge but because the structure of information geometry — specifically the Fisher-Rao metric and its curvature — is the correct description of the objective, observer-

independent geometry of the space of entropic states.

What the inversion DOES mean is that the ordering of explanation must be reversed. In the standard hierarchy, the explanatory chain runs: matter $\rightarrow$ microstates $\rightarrow$ entropy $\rightarrow$ information. In ToE, the chain runs: entropic field $\rightarrow$ information geometry $\rightarrow$ spacetime metric $\rightarrow$ matter. The physical content at each step remains; only the direction of derivation is reversed. The situation is precisely analogous to the Copernican revolution in astronomy: in the Ptolemaic system, the explanatory chain ran from the Earth outward, with all celestial motions explained as epicycles around an Earth-centered frame. In the Copernican system, the explanatory chain runs from the Sun outward. The observational facts — the positions of the planets — are the same. But the explanatory structure is reversed, and from that reversal flows an enormous simplification and a far more powerful predictive framework. ToE claims the same for the entropy-matter inversion: the observational facts of general relativity, quantum mechanics, and thermodynamics remain; only the explanatory direction is reversed; and from that reversal flows a unified account of both sides of the Einstein field equations from a single dynamical field.

One further remark: the revolutionary inversion is not an arbitrary choice. It is mandated by the results reviewed in Section 3.2 above. Jacobson's derivation of the Einstein equations from thermodynamics is not consistent with matter being prior to entropy: if the Einstein equations are an equation of state of the entropy, then entropy is prior, period. The Copernican analogy is apt here too: once Kepler's laws were established, the Copernican picture was not just simpler — it was required. Once Jacobson's result is established, ToE's inversion is not

just simpler — it is required.



Section 4. Entropy as a Field — Mathematical Formalization

4.1 What Is a Field?

Before we can say what it means for entropy to be a field, we must be clear about what fields in general are. A field is a mathematical assignment of a definite object to every point of a base space. The precise type of object assigned determines the type of field:

- A **scalar field** assigns one real number to every point of the base space. Examples: temperature $T(x)$ in a room, the gravitational potential $\varphi(x)$ in Newtonian gravity, the Higgs field $h(x)$.
- A **vector field** assigns a vector (an element of a vector space) to every point. Examples: the electric field $E(x)$, the velocity field $v(x)$ of a fluid, the four-velocity $u^\mu(x)$ of a relativistic fluid.
- A **tensor field** assigns a tensor to every point. Examples: the metric tensor $g_{\mu\nu}(x)$, the Riemann curvature tensor $R^\rho{}_{\sigma\mu\nu}(x)$, the stress-energy tensor $T_{\mu\nu}(x)$.

The entropy field $S: M \rightarrow \mathbb{R}$ assigns a real number to every point of a manifold M . This makes it a scalar field. Mathematically, it is exactly the same type of object as a temperature field or a gravitational potential field. There is nothing exotic about calling entropy a scalar field. The question is not

whether it is a field — it obviously is — but whether it is a *dynamical* field with its own equation of motion, rather than a derived quantity computed from some more fundamental field.

4.2 What Makes Entropy a Dynamical Field in ToE?

In standard thermodynamics, entropy is a state function. Given the thermodynamic state of a system (specified, for example, by its internal energy U , volume V , and particle number N), the entropy is determined: $S = S(U, V, N)$. Entropy has no independent dynamics; it is determined by the dynamics of the underlying matter.

In ToE, entropy is promoted to a local dynamical field $S(x)$ with its own equation of motion: the Master Entropic Equation (MEE). The MEE is derived from the Obidi Action A_{Obidi} by the variational principle. Just as Maxwell's equations are derived from the electromagnetic action by varying with respect to the gauge potential A_μ , and just as the Einstein field equations are derived from the Einstein-Hilbert action by varying with respect to the metric $g_{\mu\nu}$, the MEE is derived from the Obidi Action by varying with respect to the entropy field $S(x)$. The MEE will be derived in full in Part II; here we note its schematic form:

(4.1)

$$\delta A_{Obidi} /$$

δ

$$S(x) = 0$$

$\rightarrow$

MEE for $S(x)$

This is the sense in which $S(x)$ is dynamical in ToE: it satisfies a field equation that determines its evolution, just as A_μ satisfies Maxwell's equations and $g_{\mu\nu}$ satisfies Einstein's equations. The distinction between a state function and a dynamical field is precisely the distinction between a variable determined by other variables (state function) and a variable with its own independent equation of motion (dynamical field).

4.3 The Three Levels of the Entropy Field

The entropy field is a scalar field, but it gives rise to geometric objects at three distinct levels, each richer than the last:

Level 1: $S(x)$ itself — a scalar. At each point p of the manifold M , $S(p)$ is a single real number. It has no direction, no index, no transformation behavior under coordinate changes (it is invariant). This is the simplest level. Physically, $S(p)$ is the local entropy density at the point p — a measure of the informational multiplicity of microscopic configurations in an infinitesimal neighborhood of p .

Level 2: $\partial_\mu S$ — a covector field. The partial derivative of S with respect to coordinate x^μ gives the rate of change of entropy in the μ -direction. The object $\partial_\mu S = (\partial S / \partial x^\mu)$ is a covector — it transforms as a one-form under coordinate changes. It lives naturally in the cotangent space T_p^*M , not in the tangent space T_pM . This distinction is crucial and will be elaborated in Section 7.

Physically, the entropy gradient $\partial_\mu S$ drives entropic forces. Just as the gradient of potential energy drives mechanical forces, the gradient of entropy drives entropic forces. In Verlinde's picture, gravity is precisely the force arising from the gradient of entropy on a holographic screen. In ToE, $\partial_\mu S$ is the fundamental object from which all forces are derived.

Level 3: $(\nabla S)^\mu = g^{\mu\nu} \partial_\nu S$ — a vector field. Once a metric $g_{\mu\nu}$ is available, one can raise the index on the covector $\partial_\mu S$ to obtain a vector field. The vector $(\nabla S)^\mu = g^{\mu\nu} \partial_\nu S$ points in the direction of steepest entropy increase. It belongs to the tangent bundle TM . Physically, this is the entropic current — the direction and rate at which entropy "flows" in the manifold. In ToE, this vector appears in the Obidi Transformation and in the entropic geodesic equation.

The key subtlety is this: $\partial_\mu S$ (Level 2) is metric-independent and always well-defined on any manifold. $(\nabla S)^\mu$ (Level 3) requires a metric and is therefore defined only after a metric structure is established. In ToE, this means that the fundamental object is $\partial_\mu S$, and the metric $g_{\mu\nu}$ (the Obidi Metric) is derived from the information geometry. Level 3 is then constructed from Level 2 and the derived metric — not the other way around. This is why the Obidi Transformation goes from information metric to spacetime metric, not from spacetime metric to information structure.

4.4 The Observable Consequences of the Entropy Field

The entropy field is not merely conceptual. It has concrete, testable consequences for observable physics. Four are briefly noted here; their full derivation belongs to Parts II and III:

- **Entropic forces:** The gradient $\partial_\mu S$ drives forces proportional to the local entropy gradient. At the macroscopic level, gravity is the

dominant such force. The gravitational field of any massive body corresponds, in ToE, to an entropy gradient around the body — the body increases the local entropy of its neighborhood, and nearby objects move toward higher entropy (toward the body) because this is the thermodynamically favorable direction. This is Verlinde's entropic gravity, but in ToE it is derived from the Obidi Action rather than postulated.

- **Time dilation and gravitational redshift:** In ToE, the rate of a clock at a point p is determined by the local value of the entropy field $S(p)$. Points of higher entropy have slower clocks — time runs more slowly in regions of higher entropy gradient. This reproduces the gravitational time dilation of GR in the appropriate limit.
- **Mass from entropic localization:** In ToE, mass is not a primitive property of matter. It arises from the localization and self-stabilization of entropy fluctuations around a local maximum of the entropy field. A localized, stable entropy configuration has a definite rest-frame energy, which is identified with the mass of the corresponding particle. This mechanism will be developed in Part II.
- **The Entropic Speed Limit:** The propagation speed of entropy gradients is bounded above by c . This is identified in ToE (first established in Letters IIA-C) as the origin of the universal speed limit: c is the Entropic Speed Limit, the maximum rate at which entropic information can propagate through the manifold. Special relativity follows as a consequence of this limit.

Scholium 4.1 — On the Objectivity of the Entropy Field

A frequent objection to elevating entropy to a dynamical field is that

entropy is subjective — it depends on what macroscopic variables one chooses to observe and on the coarse-graining procedure used to define macrostates. This objection deserves a careful answer.

It is true that in classical statistical mechanics, the entropy of a system depends on the choice of macrostate partition: a different coarse-graining gives a different entropy value. In this sense, the standard Boltzmann entropy is coarse-graining-dependent. However, ToE does not use the Boltzmann entropy as its primary field. The entropy field $S(x)$ in ToE is the differential entropy (also called continuous entropy) of the local probability distribution over entropic microstates, evaluated in the canonical coarse-graining defined by the information-geometric structure of the statistical manifold. This canonical coarse-graining is not arbitrary: it is fixed by the Fisher-Rao geometry, which (by Čencov's theorem) is the unique geometry compatible with statistical inference. In this sense, the entropy field $S(x)$ in ToE is as objective as the electromagnetic potential $A_\mu(x)$ (which also has a gauge-dependence that must be resolved by a physical principle — gauge fixing — before it becomes a definite quantity).



PART II: THE COMPLETE MATHEMATICAL FOUNDATIONS

Section 5. Topological Spaces, Manifolds, and Charts

5.1 Topological Spaces

All of modern geometry begins with the concept of a topological space. A topological space is the minimal structure needed to make the notion of "continuity" mathematically precise, and continuity is the prerequisite for calculus, which is the language of all physical field theories.

Definition 5.1 (Topological Space). A topological space is a pair (X, τ) where X is a set and τ is a collection of subsets of X (called the *open sets*) satisfying three axioms:

1. The empty set $\emptyset$ and the whole set X are in τ .
2. Any union of elements of τ is in τ (arbitrary unions of open sets are open).
3. Any finite intersection of elements of τ is in τ (finite intersections of open sets are open).

The collection τ is called a *topology* on X . The open sets encode the "shape" of the space without reference to any metric or distance function.

Examples:

- The real line $\mathbb{R}$ with the standard topology: open sets are unions of open intervals (a, b) .
- The n -sphere $S^n = \{x \in \mathbb{R}^{n+1} \mid |x|^2 = 1\}$ with the topology inherited from $\mathbb{R}^{n+1}$.
- The torus $T^2 = S^1 \times S^1$: a circle times a circle, the surface of a donut.

Once we have a topology, we can define continuous maps: a function $f: X \rightarrow Y$ between topological spaces is continuous if the preimage of every open set in

Y is open in X . This abstract definition reduces to the familiar epsilon-delta definition of continuity when $X = Y = \mathbb{R}$.

5.2 Differentiable Manifolds

A topological space is a very coarse structure — it allows only continuous maps, not differentiable ones. To do calculus, we need more: we need the structure of a differentiable manifold.

Definition 5.2 (Differentiable Manifold). An n -dimensional differentiable manifold M is a topological space equipped with an *atlas*: a collection of pairs $\{(U_\alpha, \varphi_\alpha)\}$ called *charts*, where:

- The sets $\{U_\alpha\}$ are open subsets of M that together cover all of M : $\cup_\alpha U_\alpha = M$.
- Each $\varphi_\alpha: U_\alpha \rightarrow \mathbb{R}^n$ is a homeomorphism (a continuous bijection with continuous inverse) onto an open subset of $\mathbb{R}^n$. The map φ_α is called a *coordinate chart* or *local coordinate system*.
- Whenever two charts overlap — whenever $U_\alpha \cap U_\beta \neq \emptyset$ — the *transition map* $\varphi_\beta \circ \varphi_\alpha^{-1}: \varphi_\alpha(U_\alpha \cap U_\beta) \rightarrow \varphi_\beta(U_\alpha \cap U_\beta)$ is a smooth (infinitely differentiable, C^∞) map between open subsets of $\mathbb{R}^n$.

The coordinates $\varphi_\alpha(p) = (x^1(p), \dots, x^n(p))$ give the local coordinate description of the point $p \in U_\alpha$. Different charts give different coordinates for the same point, but the transition maps guarantee that calculus done in one chart is consistent with calculus done in another.

Standard examples relevant to ToE:

- $\mathbb{R}^n$: Euclidean n -space, with a single global chart (the identity map). This is both a Riemannian manifold (with the standard flat metric $g_{ij} = \delta_{ij}$) and the model space that all manifolds locally resemble.

- **S^2 (the 2-sphere):** Cannot be covered by a single chart (there is no smooth bijection from the whole sphere to any open subset of $\mathbb{R}^2$). The standard atlas uses two charts: stereographic projection from the north pole and from the south pole.
- **T^2 (the torus):** Can be described as the quotient $\mathbb{R}^2/\mathbb{Z}^2$ with coordinates $(\theta, \varphi) \in [0, 2\pi) \times [0, 2\pi)$.
- **The Lorentzian spacetime manifold of GR:** A 4-dimensional differentiable manifold $(M, g_{\mu\nu})$ equipped with a pseudo-Riemannian metric of signature $(-, +, +, +)$. Local coordinates are $(x^0, x^1, x^2, x^3) = (t, x, y, z)$ or (t, r, θ, φ) in spherical coordinates.
- **The statistical manifold of ToE:** A smooth manifold $\mathcal{A}$ whose points are probability distributions over entropic microstates. Its coordinates are the parameters $\theta = (\theta^i)$ labeling the distributions. This is a Riemannian manifold with the Fisher-Rao metric (3.7).

5.3 Smooth Functions on Manifolds

Once a manifold structure is in place, we can define smooth functions. The space $C^\infty(M)$ consists of all functions $f: M \rightarrow \mathbb{R}$ that are smooth — infinitely differentiable in every local coordinate chart. In local coordinates $(x^1, \dots, x^n)$, the function f is represented by a smooth function $f(x^1, \dots, x^n)$ of real variables, and the smoothness condition simply means that all partial derivatives of all orders exist and are continuous.

The space $C^\infty(M)$ is an algebra under pointwise addition and multiplication of functions. It is the fundamental algebraic object associated with a smooth manifold; in the modern algebraic approach to differential geometry, one can reconstruct the manifold from $C^\infty(M)$ alone.

For ToE, the entropy field $S: M \rightarrow \mathbb{R}$ is a smooth function in $C^\infty(M)$ — it is a smooth scalar field on the information manifold M . Its smoothness is the mathematical expression of the assumption that entropic states vary continuously and differentiably from point to point. Discontinuous entropy fields would correspond to phase transitions; the smooth field is the generic, non-critical case.

5.4 Physical Example: Gas in a Box

Consider a classical ideal gas of N particles in a box of volume V . The configuration space of the gas is the set of all possible positions and momenta of all N particles: this is the phase space $\Gamma = (\mathbb{R}^3 \times \mathbb{R}^3)^N = \mathbb{R}^{6N}$, a $6N$ -dimensional manifold. A point $\gamma \in \Gamma$ is a complete specification of the positions and momenta of all particles. This is a differentiable manifold with global coordinates (q_a^i, p_a^i) where $a = 1, \dots, N$ labels particles and $i = 1, 2, 3$ labels spatial directions.

Entropy is a function on this manifold. The Boltzmann entropy $S: \Gamma \rightarrow \mathbb{R}$ assigns to each configuration γ the logarithm of the volume of the macrostate containing γ . The entropy field of ToE generalizes this: instead of being defined on the classical phase-space manifold Γ , it is defined on the statistical manifold Λ of probability distributions over Γ , where each point of Λ specifies an entire probability distribution over microstates, not a single microstate.

Scholium 5.1 — On Why Manifolds Are the Right Setting

The reader may wonder why such abstraction is necessary. Why not work directly with coordinates, as physicists have done since Newton? The answer is that coordinate-based calculations in physics constantly generate spurious complexity: terms that look non-zero in one

coordinate system vanish in another, and it is often unclear which quantities have invariant physical meaning and which are artifacts of the coordinate choice. Manifold theory resolves this by making the coordinate-independence manifest. When we say " $S(x)$ is a scalar field on M ," we mean precisely that its numerical value at each point is coordinate-independent. When we say " $\partial_\mu S$ is a covector field," we mean precisely that it transforms in the specific way (described in Section 7) that guarantees its contraction with any vector is coordinate-independent. All of the ToE derivations are most cleanly stated in coordinate-free (intrinsic) form, and then translated to coordinates only when explicit calculations are needed. The manifold language is not decoration; it is the language in which the theory's coordinate-independence — and hence its physical objectivity — is most clearly expressed.



Section 6. Vectors, Tangent Spaces, and the Tangent Bundle

6.1 The Tangent Vector as a Derivation

In elementary physics, a vector is an arrow with magnitude and direction. On a curved manifold, this notion must be made more precise, because there is no canonical way to compare arrows at different points of a curved space. The key insight, due to Élie Cartan and others, is to define a tangent vector abstractly as an object that "differentiates functions along a curve."

Definition 6.1 (Tangent Vector). Let M be a smooth manifold and $p \in M$. A *tangent vector* at p is a linear map $X_p: C^\infty(M) \rightarrow \mathbb{R}$ that satisfies the *Leibniz rule* (product rule):

(6.1)

$$X_p(fg) = (X_p f)g(p) + f(p)(X_p g)$$

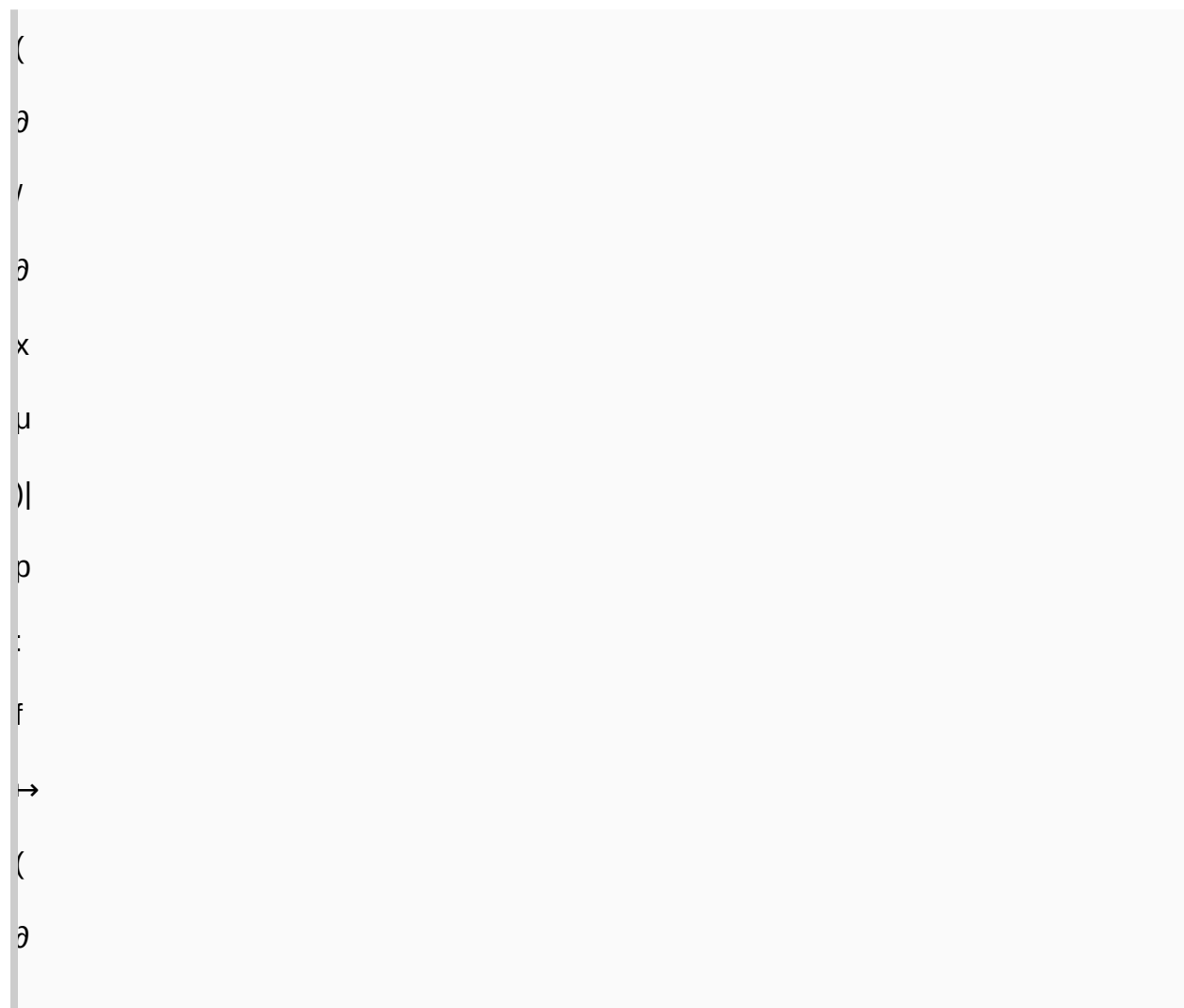
for all $f, g \in C^\infty(M)$

Such a map is called a *derivation* at p . Linearity means $X_p(\alpha f + \beta g) = \alpha X_p(f) + \beta X_p(g)$ for real numbers α, β . The Leibniz rule is the abstract version of the product rule of differentiation.

This abstract definition captures the intuitive content of "a direction and rate of change at a point": the number $X_p(f)$ is "the rate of change of f in the direction X at the point p ."

In a local coordinate system $(x^1, \dots, x^n)$ around p , the most natural tangent vectors are the *coordinate basis vectors*:

(6.2)



$f/$
 ∂
 x
 μ
 $) (p)$

These are the partial derivative operators, which are derivations at p . One can show that the n vectors $\partial/\partial x^1|_p, \dots, \partial/\partial x^n|_p$ form a basis for all tangent vectors at p , so every tangent vector can be written:

(6.3)

X
 p
 $= X$
 μ
 $($
 ∂
 $/$
 ∂
 x
 μ
 $)|$

where $X^\mu = X_p(x^\mu) \in \mathbb{R}$ are the components of X_p in this basis. The Einstein summation convention is used: repeated indices (one up, one down) are summed.

6.2 The Tangent Space

Definition 6.2 (Tangent Space). The *tangent space* $T_p M$ at a point $p \in M$ is the set of all tangent vectors at p . Under the natural operations of addition and scalar multiplication of derivations, $T_p M$ is a real vector space of dimension equal to $n = \dim M$.

The key geometric picture: $T_p M$ is the "flat approximation" to the curved manifold M at the point p . Just as the tangent plane to the Earth at your current location is the flat approximation to the Earth's curved surface, $T_p M$ is the flat approximation to M at p . But while the tangent plane in physical space is a subset of the ambient 3-space, the abstract tangent space $T_p M$ does not require any ambient space; it is intrinsically defined.

6.3 The Tangent Bundle

Definition 6.3 (Tangent Bundle). The *tangent bundle* TM is the disjoint union of all tangent spaces:

(6.4)

$TM =$

$\sqcup$

$p \in M$

T

p

$M = \{(p, v) : p$

$\in$

M, v

$\in$

T

p

$M\}$

The tangent bundle is itself a smooth manifold of dimension $2n$. It comes equipped with a natural *projection map* $\pi: TM \rightarrow M$ defined by $\pi(p, v) = p$, which sends each tangent vector to its base point. The fiber of TM above a point p is $\pi^{-1}(p) = T_p M \cong \mathbb{R}^n$.

Local coordinates on TM are given by (x^μ, v^μ) where x^μ are the base-point coordinates and v^μ are the fiber coordinates (the components of the tangent vector).

Definition 6.4 (Vector Field). A *vector field* on M is a smooth map $X: M \rightarrow TM$ satisfying $\pi \circ X = id_M$, meaning it assigns a tangent vector at each point: $X: p \mapsto (p, X_p)$. In local coordinates: $X = X^\mu(x) \partial/\partial x^\mu$ where the components $X^\mu(x)$ are smooth functions on M . A vector field is a smooth section of the tangent bundle.

6.4 Change of Basis and the Transformation Law for Vectors

When passing from one coordinate system (x^μ) to another (x'^μ) , the coordinate basis vectors transform as:

(6.5)

$$\frac{\partial}{\partial x'^\mu} = \frac{\partial x^\nu}{\partial x'^\mu} \frac{\partial}{\partial x^\nu}$$

x

v

Since a vector $X = X^\mu \partial/\partial x^\mu = X^\mu \partial/\partial x^\mu$ is a geometrically definite object (independent of coordinates), its components must transform *contravariantly*:

(6.6)

X'

μ

= (

∂

x'

μ

/

∂

x

v

) X

v

This is the *contravariant transformation law*. A collection of n functions $V^\mu(x)$ is a vector field if and only if it transforms according to (6.6) under coordinate

changes. This is how vectors are recognized in physics: by their transformation properties.

6.5 Physical Examples

- **The velocity field of a fluid:** At each point x of a 3-dimensional fluid domain, the velocity $v^i(x)$ is a vector in $T_x\mathbb{R}^3$. The collection of all these velocities is a vector field on the fluid domain, which is a section of the tangent bundle $T\mathbb{R}^3$.
- **The four-velocity in special relativity:** A relativistic particle moving through Minkowski spacetime $\mathbb{R}^{1,3}$ has a four-velocity $u^\mu = dx^\mu/d\tau$ (derivative of spacetime position with respect to proper time τ). This is a section of $T\mathbb{R}^{1,3}$ along the particle's worldline. It satisfies $g_{\mu\nu}u^\mu u^\nu = -c^2$ (normalized to -1 in natural units).
- **The entropic flow vector in ToE:** The contravariant entropy gradient $(\nabla S)^\mu = g^{\mu\nu}\partial_\nu S$ is a section of the tangent bundle TM on the information manifold M . It points in the direction of steepest entropy increase. Once the Obidi Transformation maps the information manifold to spacetime, this vector becomes a vector field on spacetime — the entropic current that sources the energy-momentum distribution of matter.

6.6 Worked Example: Tangent Vectors on S^2

On the 2-sphere S^2 with standard spherical coordinates (θ, φ) (where $\theta \in (0, \pi)$ is the polar angle and $\varphi \in [0, 2\pi)$ is the azimuthal angle), the coordinate basis at any point $p = (\theta_0, \varphi_0)$ is:

(6.7)

$$\left\{ \frac{\partial}{\partial \theta}, \frac{\partial}{\partial \varphi} \right\}$$

The tangent space $T_p\mathcal{S}^2$ is a 2-dimensional vector space spanned by these two basis vectors. A general tangent vector at p is:

(6.8)

$$X_p$$

$$\begin{aligned}
 &= a \\
 &\partial \\
 &/ \\
 &\partial \\
 &\theta \\
 &| \\
 &p \\
 &+ b \\
 &\partial \\
 &/ \\
 &\partial \\
 &\varphi \\
 &| \\
 &p
 \end{aligned}$$

for real numbers a, b . This vector describes an infinitesimal displacement from p in the direction $a\hat{\theta} + b\hat{\varphi}$ on the sphere. The action of X_p on a smooth function $f(\theta, \varphi)$ is:

(6.9)

$$\begin{aligned}
 &X \\
 &p
 \end{aligned}$$

$$(f) = a ($$

$$\partial$$

$$f/$$

$$\partial$$

$$\theta$$

$$)(p) + b ($$

$$\partial$$

$$f/$$

$$\partial$$

$$\varphi$$

$$)(p)$$

which is the directional derivative of f in the direction (a, b) at p . Note that the basis vectors $\partial/\partial\theta$ and $\partial/\partial\varphi$ are not normalized with respect to the round metric on S^2 . The metric would give $|\partial/\partial\theta| = 1$ (unit length in the θ -direction) and $|\partial/\partial\varphi| = |\sin \theta|$ (length depends on latitude — the φ -circles shrink to a point at the poles).

Scholium 6.1 — The Tangent Bundle as the Arena for Entropic Dynamics

The tangent bundle TM is the natural arena for all dynamical theories on a manifold. Classical mechanics takes place in the tangent bundle of the configuration manifold (or equivalently in the cotangent bundle, which is the phase space). General relativity's geodesic equation is a second-order ODE on TM . The entropy field's dynamical content in ToE resides partly in the tangent bundle: the entropic flow vector field $(\nabla S)^\mu$ is a

section of TM , and the entropic geodesic equation — which determines how the entropy field propagates — is an equation on TM . Understanding the tangent bundle is therefore not merely a mathematical prerequisite; it is understanding the space in which entropy flows.



Section 7. Covectors, Cotangent Spaces, and the Cotangent Bundle

7.1 What Is a Covector?

A covector (also called a one-form, a linear functional, or a covariant vector) at a point $p \in M$ is a linear map from the tangent space to the real numbers:

Definition 7.1 (Covector). A *covector* at $p \in M$ is a linear map $\omega_p: T_p M \rightarrow \mathbb{R}$. It takes a tangent vector as input and returns a real number.

The value $\omega_p(X_p) \in \mathbb{R}$ is sometimes written $\langle \omega_p, X_p \rangle$ or $\omega_p \cdot X_p$ to emphasize the pairing between covectors and vectors. This pairing is the fundamental algebraic operation in differential geometry: a one-form eats a vector and produces a number.

In local coordinates (x^μ) , the natural basis for covectors is the set of *dual basis one-forms* $\{dx^\mu\}$ defined by their action on the coordinate basis vectors:

(7.1)

dx

μ

$($

∂

$/$

∂

x

ν

$) =$

δ

μ

ν

where δ^μ_ν is the Kronecker delta (1 if $\mu = \nu$, 0 otherwise). Every covector at p can be expanded:

(7.2)

ω

p

$=$

ω

μ

dx

μ

|

p

where $\omega_\mu = \omega_p(\partial/\partial x^\mu)$ are the components of ω_p . The action of ω_p on a vector $X_p = X^\mu \partial/\partial x^\mu$ is:

(7.3)

ω

p

(X

p

) =

ω

μ

X

μ

which is the standard component-wise contraction. Note the pattern: the covector has a lower index ω_μ , the vector has an upper index X^μ , and the contraction sums over the shared index to produce a scalar.

7.2 The Cotangent Space and Cotangent Bundle

Definition 7.2 (Cotangent Space). The *cotangent space* T_p^*M at a point $p \in M$ is the vector space of all covectors at p . It is the dual space of T_pM : $T_p^*M = (T_pM)^*$. It has dimension $n = \dim M$.

Definition 7.3 (Cotangent Bundle). The *cotangent bundle* T^*M is the disjoint union of all cotangent spaces:

(7.4)

$T^*M =$

$\bigsqcup$

$p \in M$

T_p^*

p

M

It is a $2n$ -dimensional smooth manifold with projection $\pi: T^*M \rightarrow M$. A *differential one-form* is a smooth section of T^*M : a smooth map $\omega: M \rightarrow T^*M$ with $\pi \circ \omega = \text{id}_M$, assigning a covector at each point of M . In local coordinates: $\omega = \omega_\mu(x) dx^\mu$ where the components $\omega_\mu(x)$ are smooth functions.

The *exterior derivative* produces one-forms from smooth functions. For any $f \in C^\infty(M)$:

(7.5)

$df =$ (

∂

$f/$

∂

x

μ

) dx

μ

This is the gradient of f , expressed as a one-form (covector field) on M .

7.3 The Transformation Law for Covectors

Under a change of coordinates $x^\mu \rightarrow x'^\mu(x)$, the basis one-forms transform as:

(7.6)

dx'

μ

$=$ (

∂

x'

μ

$/$

∂

x

ν

) dx

ν

Since a one-form $\omega = \omega_\mu dx^\mu = \omega'_\mu dx'^\mu$ is coordinate-independent, its components must transform *covariantly*:

(7.7)

ω

ω_μ

μ

$= ($

∂

x

ν

$/$

∂

x'

μ

)

ω

v

Compare with the vector transformation law (6.6): vectors transform with $\partial x'^{\mu}/\partial x^{\nu}$ (the Jacobian), while covectors transform with $\partial x^{\nu}/\partial x'^{\mu}$ (the inverse Jacobian). Hence vectors are "contravariant" and covectors are "covariant" — they transform in opposite ways under coordinate changes, but in such a way that their inner product (7.3) remains a scalar.

7.4 The Key Point for ToE: The Entropy Gradient Is a Covector

This is a point of fundamental physical importance that must be emphasized clearly. The entropy gradient:

(7.8)

$dS =$ (

∂

$S/$

∂

x

μ

) dx

μ

is *naturally a covector field* (a one-form), not a vector field. It lives in the cotangent bundle T^*M , not the tangent bundle TM . Its components are $(\partial S/\partial x^\mu)$ with a *lower* index — the index is downstairs.

This is not just an index bookkeeping convention. It reflects a deep geometric fact: the gradient of a function is naturally a one-form because it is the object that measures the *rate of change of the function per unit displacement*. The displacement is a tangent vector (upper index); the rate of change is the inner product of the displacement with the gradient; and for this inner product to be invariant, the gradient must be a covector (lower index).

Only once a metric tensor $g_{\mu\nu}$ is available can one raise the index to obtain the entropy gradient as a vector:

(7.9)

$$\left(\nabla S \right)_\mu = g_{\mu\nu} \frac{\partial S}{\partial x^\nu} \in T_x M$$

In ToE, the metric $g_{\mu\nu}$ (the Obidi Metric) is derived from the Fisher-Rao information geometry, not postulated. This means that the fundamental quantity $\partial_\mu S$ (the covector, equation 7.8) is primary, and the vector $(\nabla S)^\mu$ (equation 7.9) is secondary — it requires the derived metric. The Obidi Transformation is the precise operation that establishes the Obidi Metric, after which the vector form of the entropy gradient becomes available.

7.5 Physical Examples

- **Pressure gradient:** The gradient of pressure $dP = (\partial P / \partial x^i) dx^i$ in a fluid is a covector. The force per unit volume on a fluid element is $f^i = -g^{ij} \partial_j P$, where the metric is needed to raise the index from covector to vector.
- **Temperature gradient:** $dT = (\partial T / \partial x^\mu) dx^\mu$ is a covector. Heat flows along the vector $q^\mu = -\kappa g^{\mu\nu} \partial_\nu T$ (Fourier's law), which again requires the metric.
- **Entropy gradient in ToE:** $dS = (\partial S / \partial x^\mu) dx^\mu$ is the fundamental covector of ToE. It drives entropic forces. Its contracted square $G^{F,\mu\nu} \partial_\mu S \partial_\nu S$ (where $G^{F,\mu\nu}$ is the inverse Fisher-Rao metric) is a scalar that appears in the Obidi Action.
- **The electromagnetic potential:** The four-potential A_μ is a covector field. The field strength tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is a (0,2) tensor formed from the exterior derivative of A .

7.6 Worked Example: Entropy Gradient in a Gaussian Model

Consider the statistical manifold of Gaussian distributions on $\mathbb{R}$ with parameters $\theta = (\mu, \sigma)$:

(7.10)

$$p(x|\mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$\sigma^2)$$

The differential entropy of this distribution is:

(7.11)

$$S(\mu, \sigma^2) = \frac{1}{2} \ln(2\pi\sigma^2) + \frac{1}{2}$$

$e) + \ln$

σ

This function does not depend on μ at all — the entropy of a Gaussian distribution is determined only by its width σ , not by where it is centered. The entropy gradient (one-form) on the 2-dimensional statistical manifold with coordinates (μ, σ) is:

(7.12)

$dS = ($

∂

$S/$

∂

μ

$) d$

μ

$+ ($

∂

$S/$

∂

σ

$) d$

$$\begin{aligned}
 & \sigma \\
 & = 0 \\
 & \cdot \\
 & d \\
 & \mu \\
 & + (1/ \\
 & \sigma \\
 &) d \\
 & \sigma \\
 & = (1/ \\
 & \sigma \\
 &) d \\
 & \sigma
 \end{aligned}$$

So $dS = (1/\sigma) d\sigma$. The entropy gradient points purely in the σ -direction. Moving the center of the distribution (changing μ) does not change entropy; only broadening or narrowing it (changing σ) does. The physical interpretation in ToE is direct: the force that acts on the entropic configuration is $\propto dS$, meaning it acts only in the σ -direction — it drives the distribution toward larger or smaller σ , but has no effect on the mean position. This corresponds to a "spreading force" that acts on the width of a statistical distribution, which is the entropic origin of certain dispersive effects in ToE.

Scholium 7.1 — Why the Distinction Between Vectors and Covectors Matters

for ToE

*In elementary physics courses, vectors and covectors are often conflated because, in Cartesian coordinates on flat Euclidean space, the metric is the identity matrix and raising or lowering indices does nothing: $V^i = \delta^{ij} V_j = V_i$. The distinction is invisible in flat space with Cartesian coordinates. But in curved space, in non-Cartesian coordinates, or in any context where the metric is non-trivial, the distinction is essential and the conflation produces errors. The entropy gradient $\partial_\mu S$ is a covector; $g^{\mu\nu} \partial_\nu S$ is the corresponding vector; and they are different objects unless the metric is trivial. ToE's derivation of the Obidi Metric from the Fisher-Rao metric involves precisely the operation of converting information about covectors (the entropy gradient, which lives in T^*M) into information about the metric, and then using the metric to produce the corresponding vector (which lives in TM). Getting this right requires careful attention to the vector/covector distinction at every step.*

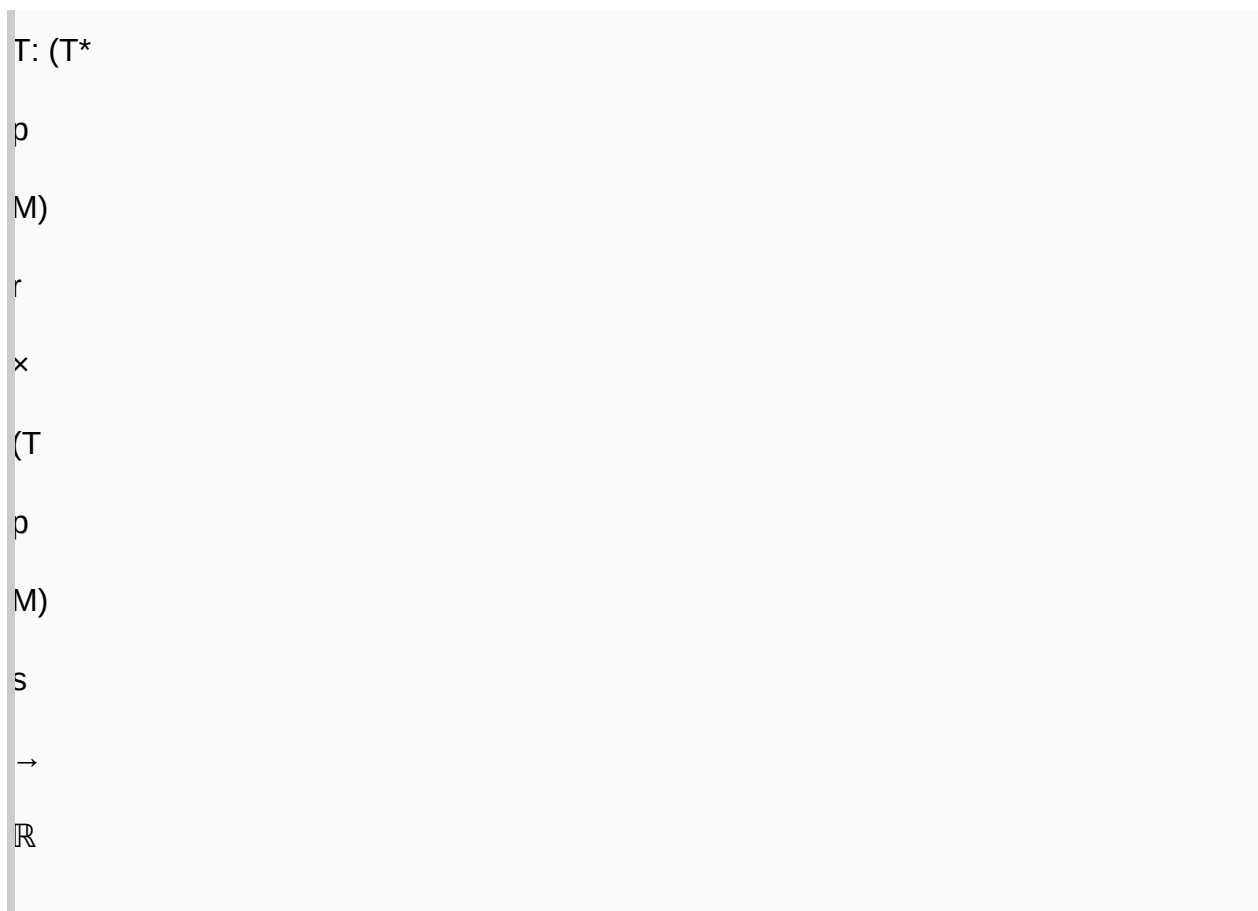


Section 8. Tensors and Tensor Fields

8.1 What Is a Tensor?

Definition 8.1 (Tensor). A *tensor of type (r, s)* at a point $p \in M$ is a multilinear map:

(8.1)



It takes r covectors and s vectors as input and produces a real number. "Multilinear" means linear in each argument separately. In local coordinates, a type (r, s) tensor has components $T^{\mu_1 \dots \mu_r}_{\nu_1 \dots \nu_s}$ where upper indices correspond to covector slots and lower indices to vector slots.

8.2 Important Special Cases

The zoo of tensors in physics becomes manageable when organized by type:

Type (r,s)	Name	Index Pattern	Physical/ToE Example
(0,0)	Scalar	No indices	Entropy density $S(x)$, Ricci scalar R
(1,0)	Contravariant	V^μ	Four-velocity u^μ ,

Type (r,s)	Name	Index Pattern	Physical/ToE Example
	vector		entropic current $(\nabla S)^\mu$
(0,1)	Covariant vector (covector)	V_μ	Entropy gradient $\partial_\mu S$, EM potential A_μ
(0,2)	Covariant 2-tensor	$T_{\mu\nu}$	Metric $g_{\mu\nu}$, Fisher-Rao metric G^F_{AB} , stress-energy $T_{\mu\nu}$
(2,0)	Contravariant 2-tensor	$T^{\mu\nu}$	Inverse metric $g^{\mu\nu}$, stress-energy $T^{\mu\nu}$
(1,1)	Mixed 2-tensor	$T^\mu{}_\nu$	Ricci tensor $R^\mu{}_\nu$, identity $\delta^\mu{}_\nu$
(0,4)	Covariant 4-tensor	$T_{\mu\nu\rho\sigma}$	Riemann curvature tensor $R_{\mu\nu\rho\sigma}$

8.3 Tensor Products, Contractions, and Index Operations

Given a type (r,s) tensor T and a type (r',s') tensor U , their *tensor product* $T \otimes U$ is a type $(r+r', s+s')$ tensor. In components:

$$(8.2)$$

(T

⊗

U)

μ

1

...

μ

r

μ

r

1

...

μ

r

r'

v

1

...

v

s

v

r

1

...

v

r

s'

= T

μ

1

...

μ

r

v

1

...

v

s

U

μ

r

1

...

μ

r

r'

v

r

1

...

ν

ρ

s'

Contraction sets one upper and one lower index equal and sums over it, reducing the tensor type by (1,1). For example, contracting the Riemann tensor $R^\rho_{\sigma\mu\nu}$ on the first and third indices gives the Ricci tensor:

(8.3)

R

$\mu\nu$

$= R$

ρ

$\mu\rho\nu$

$=$

Σ

ρ

R

ρ

$\mu\rho\nu$

Raising and lowering indices with the metric: for any vector V^μ , the metric lowers the index to produce a covector $V_\mu = g_{\mu\nu} V^\nu$. For any covector ω_μ , the inverse metric raises the index: $\omega^\mu = g^{\mu\nu} \omega_\nu$. The same operations apply to any index of any tensor.

8.4 The Metric Tensor

The metric tensor $g_{\mu\nu}$ is a symmetric, non-degenerate (0,2) tensor field. It is the fundamental geometric object on a (pseudo-)Riemannian manifold. Its role:

- **It measures lengths:** The squared length of an infinitesimal displacement dx^μ is $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$.
- **It measures angles:** The angle between vectors X and Y is determined by $g_{\mu\nu} X^\mu Y^\nu$.
- **It defines volumes:** The invariant volume element on an n -dimensional manifold is $\sqrt{|g|} dx^1 \dots dx^n$ where $|g| = |\det(g_{\mu\nu})|$.
- **It raises and lowers indices:** as described above.

Riemannian vs. Lorentzian metrics: A metric is *Riemannian* if it is positive definite: $g_{\mu\nu} X^\mu X^\nu > 0$ for all non-zero vectors X . It is *Lorentzian* (pseudo-Riemannian) if it has signature $(-, +, \dots, +)$: one negative eigenvalue and $n-1$ positive ones. The Fisher-Rao metric is Riemannian (positive definite). The Obidi Metric — obtained from the Fisher-Rao metric by the Obidi Transformation — is Lorentzian (pseudo-Riemannian), as required for its identification with the spacetime metric of general relativity. The mechanism by which a Riemannian metric becomes Lorentzian under the Obidi Transformation is one of the most important results of the theory and will be detailed in Part II.

8.5 The Key Tensors of ToE

The following tensors play central roles in ToE and will appear repeatedly in Parts II and III:

The Fisher-Rao metric G^F_{AB} : a positive-definite, symmetric (0,2) tensor on the statistical manifold $\mathcal{A}$. Indices A, B label coordinates on $\mathcal{A}$. It measures the statistical distinguishability of nearby probability distributions. Defined by:

(8.4)

$$G^F_{AB} = \int \frac{\partial p(x|\theta)}{\partial \theta^A} \frac{\partial p(x|\theta)}{\partial \theta^B} dx$$

∂

B

$\ln p(x|$

θ

$) dx$

where $\partial_A = \partial/\partial\theta^A$. This is the fundamental metric of information geometry.

The Obidi Metric $\tilde{G}_{AB}$: a Lorentzian (0,2) tensor on the information manifold, obtained from G^F_{AB} by the Obidi Transformation. It carries signature $(-,+, \dots, +)$ and is identified with the physical spacetime metric after coarse-graining via the Vuli-Ndlela Integral.

The Ricci tensor $R_{\mu\nu}$: a symmetric (0,2) tensor measuring the average curvature in planes containing each direction. It is the fundamental curvature object appearing on the left-hand side of the Einstein field equations.

The Einstein tensor $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$: a symmetric (0,2) tensor that is automatically divergence-free: $\nabla^\mu G_{\mu\nu} = 0$ by the contracted Bianchi identity. This is the left-hand side of the Einstein field equations.

The Entropic Stress-Energy Tensor $T^{ent}_{\mu\nu}$: a symmetric (0,2) tensor derived from the Obidi Action by variation with respect to the metric. It is the right-hand side of the entropic field equations and is identified, in the near-equilibrium limit, with $(8\pi G/c^4)^{-1}$ times the Einstein tensor of the emergent spacetime. Its explicit form is:

(8.5)

$$T_{\mu\nu} = -\frac{2}{\sqrt{|g|}} \frac{\delta S}{\delta g^{\mu\nu}}$$

This is the standard Hilbert definition of the stress-energy tensor from an action principle. Its explicit component form in terms of the entropy field will be given in Part II.

8.6 Symmetric and Antisymmetric Tensors

A (0,2) tensor $T_{\mu\nu}$ is *symmetric* if $T_{\mu\nu} = T_{\nu\mu}$ and *antisymmetric* if $T_{\mu\nu} = -T_{\nu\mu}$. Any (0,2) tensor decomposes uniquely:

(8.6)

T

$\mu\nu$

$= T$

$(\mu\nu)$

$+ T$

$[\mu\nu]$

where

T

$(\mu\nu)$

$=$

$\frac{1}{2}$

$(T$

$\mu\nu$

$+ T$

$\nu\mu$

$),$

T

$[\mu\nu]$

$=$

$\frac{1}{2}$

$(T$

$\mu\nu$

—

T

$\nu\mu$

)

In ToE, the metric, the Fisher-Rao metric, the Obidi Metric, the Ricci tensor, the Einstein tensor, and the Entropic Stress-Energy Tensor are all symmetric. The Riemann tensor has both symmetric and antisymmetric index structures as detailed in Section 13.

Scholium 8.1 — The Entropic Stress-Energy Tensor as a Second Moment

It is worth emphasizing at this point a deep structural parallel that will be fully developed in Section 14. The stress-energy tensor of a classical relativistic gas is computed in kinetic theory as the second moment of the momentum distribution function: $T^{\mu\nu} = \int p^\mu p^\nu f(x,p) d\mu_p$, where the integral is over momentum space (the fiber above each spacetime point). This is precisely a fiber integral of a second-moment integrand over the momentum space fiber. In ToE, the Entropic Stress-Energy Tensor $T^{\text{ent}}_{\mu\nu}$ arises by an analogous fiber integration of a second moment of the entropic momentum distribution over the fiber of the statistical manifold bundle above each spacetime point. This structural parallel is not accidental; it reflects the deep connection between kinetic theory and information geometry that ToE makes explicit. The stress-energy tensor is always a second moment; in standard physics, it is a second moment of

a distribution of matter; in ToE, it is a second moment of a distribution of entropy.



Section 9. Fiber Bundles — The Complete Theory

9.1 Definition and Intuition

The concept of a fiber bundle generalizes both the tangent bundle and the cotangent bundle, and lies at the heart of modern differential geometry, gauge theory, and the architecture of ToE. The key insight is that a fiber bundle is a space that locally looks like a product of two spaces but may be globally "twisted." The local product structure allows calculus; the global twist encodes topological and geometric information.

Definition 9.1 (Fiber Bundle). A *fiber bundle* is a tuple (E, B, π, F) where:

- E is the *total space* (the "big" space).
- B is the *base space* (the "small" space, parametrizing the fibers).
- $\pi: E \rightarrow B$ is the *projection map* (a surjective continuous map).
- F is the *typical fiber* (the space that sits above each point of B : $\pi^{-1}(b) \cong F$ for all $b \in B$).
- *Local triviality*: for every point $b \in B$, there exists an open neighborhood $U_b \subset B$ and a homeomorphism $\varphi: \pi^{-1}(U_b) \rightarrow U_b \times F$ such that $\pi = pr_1 \circ \varphi$ (where $pr_1: U_b \times F \rightarrow U_b$ is the projection onto the first factor).

The local triviality condition says: locally, the total space E looks exactly like a product $U_b \times F$. Globally, however, the fibers may be attached to the base in a "twisted" way that cannot be globally untwisted.

Geometrically: imagine a cylinder formed by stacking circles (fibers S^1) above each point of a line segment (base $B = [0,1]$). This is a trivial bundle: the total space is $E = [0,1] \times S^1$. Now imagine "twisting" the attachment as you go around a circle — after going all the way around, the fiber is attached with a twist. This gives a Möbius band, which is a non-trivial bundle with $B = S^1$ and fiber $F = [-1,1]$. The Möbius band cannot be flattened into a product without tearing; this non-triviality is its topology.

9.2 Key Examples

Bundle	Base B	Fiber F	Total Space E	Trivial?
Cylinder	S^1	$\mathbb{R}$	$S^1 \times \mathbb{R}$	Yes
Möbius band	S^1	$[-1,1]$	Twisted $[-1,1]$ -bundle	No
Tangent bundle TM	M (any manifold)	$\mathbb{R}^n$	TM	Generally No
Cotangent bundle T*M	M	$(\mathbb{R}^n)^*$	T*M	Generally No
Frame bundle	M	$GL(n,\mathbb{R})$	FM	Generally No
Statistical manifold bundle (ToE)	Spacetime M	Fiber of entropic states Λ_p	Λ	Determined by entropy dynamics

9.3 Vector Bundles

Definition 9.2 (Vector Bundle). A *vector bundle* of rank k over B is a fiber bundle $(E, B, \pi, \mathbb{R}^k)$ where the fiber is a k -dimensional real vector space, and the transition functions are linear maps (elements of $GL(k, \mathbb{R})$).

The tangent bundle TM is a vector bundle of rank n over M . The cotangent bundle T^*M is a vector bundle of rank n . More generally, any tensor bundle:

(9.1)

$$T^r_s M = \underbrace{(TM \otimes \cdots \otimes TM)}_r \otimes \underbrace{(T^*M \otimes \cdots \otimes T^*M)}_s$$

is a vector bundle of rank n^{r+s} . Sections of $T^r_s M$ are tensor fields of type (r,s) .

9.4 Sections of a Fiber Bundle

Definition 9.3 (Section). A *section* of a fiber bundle (E, B, π, F) is a smooth map $\sigma: B \rightarrow E$ satisfying $\pi \circ \sigma = id_B$. A section assigns a point in the fiber $\pi^{-1}(b) \cong F$ to each point $b \in B$.

In the physical language: a section selects one "element of the fiber" at each point of the base. Smooth sections of TM are vector fields; smooth sections of T^*M are one-forms; smooth sections of the tensor bundle $T_s^r M$ are tensor fields of type (r,s) .

9.5 Principal Bundles and Gauge Theory

Definition 9.4 (Principal G-Bundle). A *principal bundle* with structure group G (a Lie group) is a fiber bundle (P, B, π, G) where G acts freely and transitively on each fiber $\pi^{-1}(b)$ from the right. The frame bundle FM is the principal $GL(n, \mathbb{R})$ -bundle over a manifold M , consisting of all ordered bases (frames) for the tangent spaces.

Principal bundles are the mathematical foundation of gauge theories. In the electromagnetic field, the gauge group is $U(1)$ and the gauge field is a connection on a principal $U(1)$ -bundle. In the Standard Model, the gauge groups are $SU(3) \times SU(2) \times U(1)$. In GR, the frame bundle with structure group $SO(3,1)$ (the local Lorentz group) is the relevant principal bundle. In ToE, the relevant structure group is determined by the symmetries of the Obidi Metric and the Obidi Action; its identification is part of the research program reported in Parts II and III.

9.6 Connections on Fiber Bundles

A fiber bundle by itself does not specify how to "move" between fibers as one moves along the base. A connection provides this structure.

Definition 9.5 (Connection on a Vector Bundle). A *connection* on a vector bundle E over B is a bilinear map $\nabla: \Gamma(TB) \times \Gamma(E) \rightarrow \Gamma(E)$, written $(X, s) \mapsto \nabla_X s$, satisfying:

- Linearity over $C^\infty(B)$ in the first argument: $\nabla_{fX}s = f \nabla_X s$.
- The Leibniz rule in the second argument: $\nabla_X(fs) = X(f)s + f \nabla_X s$.

Here $\Gamma(TB)$ denotes the space of vector fields on B and $\Gamma(E)$ the space of sections of E . The quantity $\nabla_X s$ is the "covariant derivative of the section s in the direction X ." This generalizes the directional derivative of functions to sections of non-trivial bundles.

In local coordinates, a connection on a vector bundle is specified by connection coefficients (gauge fields in the physics language). For the tangent bundle TM , the connection coefficients are the Christoffel symbols $\Gamma^\rho_{\mu\nu}$ of the Levi-Civita connection (Section 12).

9.7 Why Fiber Bundles Matter for ToE: The Full Picture

Fiber bundles are not peripheral to ToE; they are its architectural foundation. Let us state precisely why:

(i) The statistical manifold bundle. In ToE, the fundamental object is the statistical manifold Λ whose points are probability distributions over entropic microstates. Physical spacetime M is envisioned as the base of a fiber bundle $(\Lambda, M, \pi, \Lambda_p)$ where the fiber $\Lambda_p = \pi^{-1}(p)$ above each spacetime point p is the local statistical manifold of microscopic entropic states at p . The total space Λ is the microscopic theory; the base M is the macroscopic spacetime; and the fiber Λ_p encodes all the microscopic entropy fluctuations consistent with the macroscopic entropy value $S(p)$.

(ii) Coarse-graining as fiber integration. The passage from the microscopic theory (on $\mathcal{A}$) to the macroscopic theory (on $\mathcal{M}$) is accomplished by integrating out the fiber degrees of freedom — the fiber integral, defined precisely in Section 11. This is mathematically the same operation as integrating out momentum space in kinetic theory: the macroscopic stress-energy tensor $T_{\mu\nu}$ is the fiber integral of the microscopic momentum distribution $p^\mu p^\nu f$ over the momentum fiber. In ToE, the macroscopic Entropic Stress-Energy Tensor is the fiber integral of the microscopic second moment of the entropic momentum distribution over the statistical fiber. This connection is not an analogy; it is a theorem, and its proof is given in Part II.

(iii) The Obidi Transformation as a bundle map. The Obidi Transformation can be formulated precisely as a bundle map from the statistical manifold bundle $(\mathcal{A}, \mathcal{M}, \pi, \mathcal{A}_p)$ to the tangent bundle $(TM, \mathcal{M}, \pi', \mathbb{R}^4)$ that converts the Riemannian fiber metric (Fisher-Rao) into the Lorentzian base metric (Obidi Metric). This reformulation makes the coordinate-independence of the transformation manifest.

(iv) The HMAS as a hybrid bundle. The Hybrid Metric-Affine Space (HMAS) of ToE is a specific fiber bundle structure where the base is the classical spacetime manifold $\mathcal{M}$, one fiber sector carries the Fisher-Rao geometry (classical statistical manifold), and another fiber sector carries the Fubini-Study geometry (quantum statistical manifold). The two sectors are coupled through the Obidi Metric at their boundary. The HMAS is thus a bundle with a hybrid fiber $F = F_{\text{classical}} \cup F_{\text{quantum}}$, and the classical-quantum boundary is a co-dimension-one surface in the fiber determined by the Obidi Curvature Invariant.

Scholium 9.1 — On the Universality of the Fiber Bundle Picture

The reader who knows gauge theory will recognize that the fiber bundle is the universal language in which all fundamental interactions are currently written. Electromagnetism is a connection on a $U(1)$ principal bundle. The weak force is a connection on an $SU(2)$ bundle. The strong force is a connection on an $SU(3)$ bundle. General relativity, in the first-order (Palatini) formulation, is a connection on the frame bundle with structure group $SO(3,1)$. What ToE adds to this picture is a new type of bundle — the statistical manifold bundle — with a new type of connection — the Amari-Čencov α -connection — and a new fundamental field — the entropy field $S(x)$ as a section of the trivial line bundle over the base. The fiber bundle language unifies ToE with the prior gauge-theoretic formulation of physics, while the specific new structures introduced (the statistical fiber, the information metric, the Obidi Transformation) constitute its novel physical content. To understand ToE fully is to understand how a fiber bundle with an information-geometric fiber and an entropic section reproduces, in the appropriate limits, all the gauge bundles of the Standard Model and the frame bundle of GR.



Section 10. Probability Distributions as Geometric Objects

10.1 Discrete Distributions and the Simplex

For a discrete outcome space $X = \{x_1, \dots, x_n\}$, a probability distribution is a vector $p = (p_1, \dots, p_n)$ satisfying $p_i \geq 0$ and $\sum_i p_i = 1$. The set of all such vectors is the $(n-1)$ -simplex:

(10.1)

$$\Delta^{n-1} = \{(p_1, \dots, p_n) \in \mathbb{R}^n \mid p_i \geq 0, \sum_{i=1}^n p_i = 1\}$$

$$\sum_{i=1}^n p_i = 1\}$$

This is a compact $(n-1)$ -dimensional manifold with boundary. Points in the interior have all $p_i > 0$; boundary points have some $p_i = 0$ (corresponding to degenerate distributions that assign zero probability to some outcomes). For the applications in ToE, we work with the interior of the simplex (all probabilities strictly positive), which is a smooth manifold without boundary.

For a continuous outcome space with probability densities $p(x)$, the "simplex" is replaced by the space of all smooth probability densities satisfying $\int p(x) dx = 1$, which is an infinite-dimensional manifold. ToE works with finite-dimensional parametric subfamilies of this space, called statistical manifolds.

10.2 Parametric Families and Statistical Manifolds

Definition 10.1 (Statistical Manifold). A *statistical manifold* is a smooth manifold Λ together with a smooth map $p: X \times \Lambda \rightarrow [0, \infty)$, written $p(x/\theta)$, such that for each $\theta \in \Lambda$, the function $x \mapsto p(x/\theta)$ is a probability distribution on X . The parameters $\theta = (\theta^1, \dots, \theta^k)$ are local coordinates on Λ .

The statistical manifold is simultaneously a parameter space (a set of labels for probability distributions) and a geometric space (equipped with the Fisher-Rao metric). Each point $\theta \in \Lambda$ is both a parameter value and a point on a Riemannian manifold.

10.3 Exponential Families

The most important class of statistical manifolds consists of the *exponential families*:

Definition 10.2 (Exponential Family). A parametric family $\{p(x|\theta)\}$ is an *exponential family* if it can be written:

(10.2)

$$p(x|\theta) = \exp\left(\sum_{i=1}^n \theta_i T_i(x) - A(\theta)\right) h(x)$$

where $T_i(x)$ are the *sufficient statistics*, $A(\theta)$ is the *log-partition function* (also called the cumulant generating function), and $h(x)$ is the base measure. The parameters $\theta = (\theta^i)$ are called the *natural parameters*.

The log-partition function is defined by the normalization condition:

(10.3)

$$A(\theta) = \ln \int \exp(\theta^T T(x) + h(x)) \, dx$$

Key examples relevant to ToE:

Distribution	Natural parameters θ	Sufficient statistics $T(x)$	Log-partition $A(\theta)$
Gaussian $N(\mu, \sigma^2)$	$(\mu/\sigma^2, -1/(2\sigma^2))$	(x, x^2)	$-\mu^2/(2\sigma^2) + \frac{1}{2}\ln(2\pi\sigma^2)$
Poisson $\text{Pois}(\lambda)$	$\ln \lambda$	x	$e^\theta = \lambda$
Bernoulli $\text{Ber}(q)$	$\ln(q/(1-q))$	x	$\ln(1 + e^\theta)$
Exponential $\text{Exp}(\lambda)$	$-\lambda$	x	$-\ln(-\theta)$

The log-partition function $A(\theta)$ is convex (positive semi-definite Hessian), and its second derivatives give the covariance structure of the sufficient statistics:

(10.4)

$$\frac{\partial^2 A(\theta)}{\partial \theta_i \partial \theta_j} = \text{Cov}(T_i(x), T_j(x)) = \langle T_i T_j \rangle - \langle T_i \rangle \langle T_j \rangle$$

$\langle$
 T
 i
 $\rangle$
 $\langle$
 T
 i
 $\rangle$

For an exponential family, the Fisher-Rao metric (8.4) simplifies to:

(10.5)

G
 F
 ij
 $($
 θ
 $) =$
 ∂
 i
 ∂
 i

$A(\theta)$

the Hessian of the log-partition function. This remarkable fact — that the information metric is the Hessian of a convex function — is the foundation of the dually flat geometry of exponential families, a central topic in information geometry.

10.4 Entropy of a Distribution

The differential entropy of a continuous probability distribution $p(x)$ is:

(10.6)

$$H(p) = - \int p(x) \ln p(x) dx$$

For a Gaussian $N(\mu, \sigma^2)$:

(10.7)

$$H(N(\mu, \sigma^2)) = \frac{1}{2} \ln(2\pi e \sigma^2)$$

$$\begin{aligned}
 & \cdot \\
 & \sigma \\
 & 2 \\
 &)) = \\
 & \frac{1}{2} \\
 & \ln(2 \\
 & \pi \\
 & e \\
 & \sigma \\
 & 2 \\
 &)
 \end{aligned}$$

which depends only on σ , not on μ , as we saw in Section 7.6. For a multivariate Gaussian $N(\mu, \Sigma)$ on $\mathbb{R}^n$:

(10.8)

$$\begin{aligned}
 & H(N(\\
 & \mu \\
 & , \\
 & \Sigma \\
 &)) = \\
 & \frac{1}{2}
 \end{aligned}$$

constraints at x . This ensemble is described by a probability distribution $p(y/x)$ where y labels the microscopic state and x is the spacetime point (the label of the macro-region).

As the macroscopic conditions vary smoothly from point to point — as x varies over spacetime — the local distribution $p(y/x)$ varies smoothly. The local entropy:

(10.9)

$$S(x) = - \int p(y/x) \ln p(y/x) dy$$

then becomes a smooth function of x — a scalar field on spacetime. This is the entropy field. It encodes, at each spacetime point, the informational multiplicity of microscopic configurations consistent with the macroscopic constraints there. Regions where matter is concentrated have high multiplicity (many equivalent microscopic arrangements), hence high local entropy $S(x)$. Vacuum regions have lower entropy. The gradient $\partial_\mu S$ drives entropic forces toward higher entropy — which, in the ToE picture, is the origin of gravity.

Scholium 10.1 — On the Relationship Between the Entropy Field and Thermodynamics

The reader familiar with thermodynamics will notice that equation (10.9) looks like the standard definition of entropy in statistical mechanics, but with x as a spatial label instead of a thermodynamic parameter. This is correct, and it is precisely the connection that ToE exploits. In standard thermodynamics, one considers a single system in a single macrostate and computes its entropy as a global quantity. In ToE, one imagines resolving the system into infinitesimal volume elements, computing the entropy of each volume element's local distribution, and treating the resulting function $S(x)$ as a field. This is the standard move from global thermodynamics to local thermodynamics — from entropy as a state function to entropy as a field. The novelty of ToE is not this move (which is standard in non-equilibrium thermodynamics), but the subsequent step: treating this local entropy field as the primary dynamical object and deriving everything else from it.



Section 11. Differential Forms and Integration on Manifolds

11.1 Differential k-Forms

Differential forms generalize the one-forms (covectors) of Section 7 to multi-covector objects. They are the natural objects that can be integrated on manifolds.

Definition 11.1 (Differential k-Form). A *differential k-form* on a manifold M is a completely antisymmetric $(0,k)$ tensor field. In local coordinates:

(11.1)

$$\omega = \omega_{\mu_1 \dots \mu_k}(x) dx^{\mu_1} \wedge \dots \wedge dx^{\mu_k}$$

where the *wedge product* $\wedge$ is antisymmetric: $dx^\mu \wedge dx^\nu = -dx^\nu \wedge dx^\mu$, and the components $\omega_{\mu_1 \dots \mu_k}$ are smooth functions antisymmetric in all indices. The space of k -forms on M is denoted $\Omega^k(M)$.

Special cases: $\Omega^0(M) = C^\infty(M)$ (functions), $\Omega^1(M)$ (one-forms as in Section 7), $\Omega^n(M)$ (top-degree forms, used for integration).

11.2 The Exterior Derivative

Definition 11.2 (Exterior Derivative). The *exterior derivative* $d: \Omega^k(M) \rightarrow \Omega^{k+1}(M)$ is the unique linear map satisfying:

- On 0-forms (functions): $df = (\partial f / \partial x^\mu) dx^\mu$ (the gradient).
- The Leibniz rule: $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$ for a k -form ω .
- Nilpotency: $d^2 = d \circ d = 0$.

On a 1-form $\omega = \omega_\mu dx^\mu$:

(11.2)

$$d\omega = \left(\frac{\partial \omega_\nu}{\partial x^\mu} - \omega_\nu \frac{\partial x^\mu}{\partial x^\nu} \right) dx^\mu \wedge dx^\nu$$

$$\begin{aligned}
 & \partial \\
 & \nu \\
 & \omega \\
 & \mu \\
 &) \, dx \\
 & \mu \\
 & \wedge \\
 & dx \\
 & \nu
 \end{aligned}$$

This is the curl of ω . The nilpotency $d^2 = 0$ encodes the Poincaré lemma and the Bianchi identity: in classical terms, the "curl of a gradient is zero" and "div of curl is zero." In ToE, the nilpotency of d applied to the entropy gradient dS gives $d(dS) = 0$, which is the entropic integrability condition.

11.3 Integration of n-Forms on Manifolds

On an n -dimensional oriented manifold M , a top-degree form $\omega \in \Omega^n(M)$ can be integrated. In local coordinates:

(11.3)

$$\begin{aligned}
 & \int \\
 & M \\
 & \omega
 \end{aligned}$$

$$= \int_U \omega_{1\dots n}(x) \, dx_1 \dots dx_n$$

where $U \subset \mathbb{R}^n$ is the coordinate domain. This integral is well-defined and coordinate-independent for any compactly supported top-degree form on an oriented manifold.

11.4 Stokes' Theorem

Theorem 11.1 (Stokes' Theorem). Let M be an oriented n -dimensional manifold with boundary ∂M , and let $\omega \in \Omega^{n-1}(M)$ be a smooth $(n-1)$ -form with compact support. Then:

(11.4)

$$\int_{\partial M} \omega = \int_M d\omega$$

$$\int_M \omega = \int_{\partial M} \omega$$

This single formula encompasses all of vector calculus: Green's theorem ($n=2$), the divergence theorem ($n=3$, using the Hodge dual), and the classical Stokes' theorem ($n=3$, on surfaces). In the language of physics, Stokes' theorem is the coordinate-free statement of Gauss's law and its generalizations.

In ToE, Stokes' theorem appears in the derivation of conservation laws from the action principle: varying the Obidi Action and integrating by parts produces boundary terms that vanish by Stokes' theorem if the variation decays at infinity, leaving the Euler-Lagrange equations (the MEE) as the bulk equation of motion.

11.5 Volume Forms and the Metric

Given a metric $g_{\mu\nu}$ on an n -manifold, the natural *volume form* is:

(11.5)

$$\varepsilon = \sqrt{|g|} dx^1 \wedge \dots \wedge dx^n$$

$|g| \, dx$

1

$\wedge$

dx

2

$\wedge$

...

$\wedge$

dx

n

where $|g| = |\det(g_{\mu\nu})|$. Integration of a scalar function f against this volume form gives:

(11.6)

$\int$

M

f

ε

$=$

$\int$

M

$f(x)$

$\sqrt$

$|g| dx$

1

$\dots dx$

n

This is the coordinate-invariant volume integral. In GR, the Einstein-Hilbert action is:

(11.7)

A

EH

$= (1/(16$

π

$G))$

$\int$

M

R

$\sqrt$

$|g| d$

4

x

In ToE, the Obidi Action has a similar structure but with the Ricci scalar R replaced by a functional of the entropy field and the Fisher-Rao metric, integrated against the entropic volume form. The precise form will be given in Part II.

11.6 Fiber Integrals: The Mathematics of Coarse-Graining

This subsection establishes the mathematical formalism of fiber integration, which is the precise realization of "coarse-graining" in ToE.

Definition 11.3 (Fiber Integral). Let $\pi: E \rightarrow B$ be a fiber bundle with m -dimensional fiber F and n -dimensional base B . The *fiber integral* (also called integration along the fiber or pushforward) of a k -form ω on E is a $(k-m)$ -form $\pi_*\omega$ on B defined by:

(11.8)

(

π

*

ω

)(X

1

, ..., X

$k-m$

$\gamma(b) =$

$\int$

π

-1

(b)

$\int$

X

$\circ \sim$

1

$\dots \iota;$

X

$\circ \sim$

$k-m$

ω

where $\tilde{X}_i$ are arbitrary lifts of X_i to horizontal vector fields on E , and ι_X denotes interior multiplication (contraction with X). The integral is over the fiber $\pi^{-1}(b) \cong F$.

In the simplest terms: a fiber integral takes a form defined on the "large space" E (which has both base and fiber directions) and, by integrating away the fiber directions, produces a form on the "small space" B (which has only base directions). It is the formal mathematical operation of "integrating out" the hidden degrees of freedom.

The kinetic theory example: In relativistic kinetic theory, the stress-energy tensor is computed as a fiber integral:

(11.9)

$$T_{\mu\nu}(x) = \int_{T_x^*M} p_\mu p_\nu f(x, p) \frac{d^3p}{p^0}$$

where the integral is over the momentum fiber T_x^*M (the cotangent space at x , which is the momentum space at that point), $f(x, p)$ is the one-particle distribution function on phase space, and the measure $(-g)^{1/2} d^3p/p^0$ is the Lorentz-invariant momentum-space measure. This is a fiber integral of the second moment $p^\mu p^\nu f$ over the cotangent fiber, producing a (2,0) tensor (the stress-energy tensor) on the base spacetime M .

In ToE, this formula is replaced by:

(11.10)

T
ent
μν
(x) =
[
Λ
x
q
μ
q
ν
f

ent

(x, q) d

μ

F

(q)

where the integral is over the statistical fiber Λ_x (the local statistical manifold at x), $q_\mu = \partial_\mu S$ is the entropic momentum (the entropy gradient), $I^{ent}(x, q)$ is the entropic distribution function derived from the Obidi Action, and $d\mu_F(q)$ is the Fisher-Rao volume measure on the statistical fiber. This is the fiber integral that produces the Entropic Stress-Energy Tensor from the microscopic information-geometric dynamics — equation (11.10) is the ToE generalization of the kinetic theory formula (11.9).

11.7 Pullback and Pushforward

The pullback and pushforward operations are the covariant-geometric description of change of variables. If $\varphi: M \rightarrow N$ is a smooth map, then:

The *pullback* $\varphi^*: \Omega^k(N) \rightarrow \Omega^k(M)$ pulls a k-form on N back to a k-form on M . In components, for a (0,2) tensor:

(11.11)

(

φ

$^*h)$

$\mu\nu$

$(x) = ($

∂

φ

A

/

∂

x

μ

$) ($

∂

φ

B

/

∂

x

ν

$) h$

AB

$($

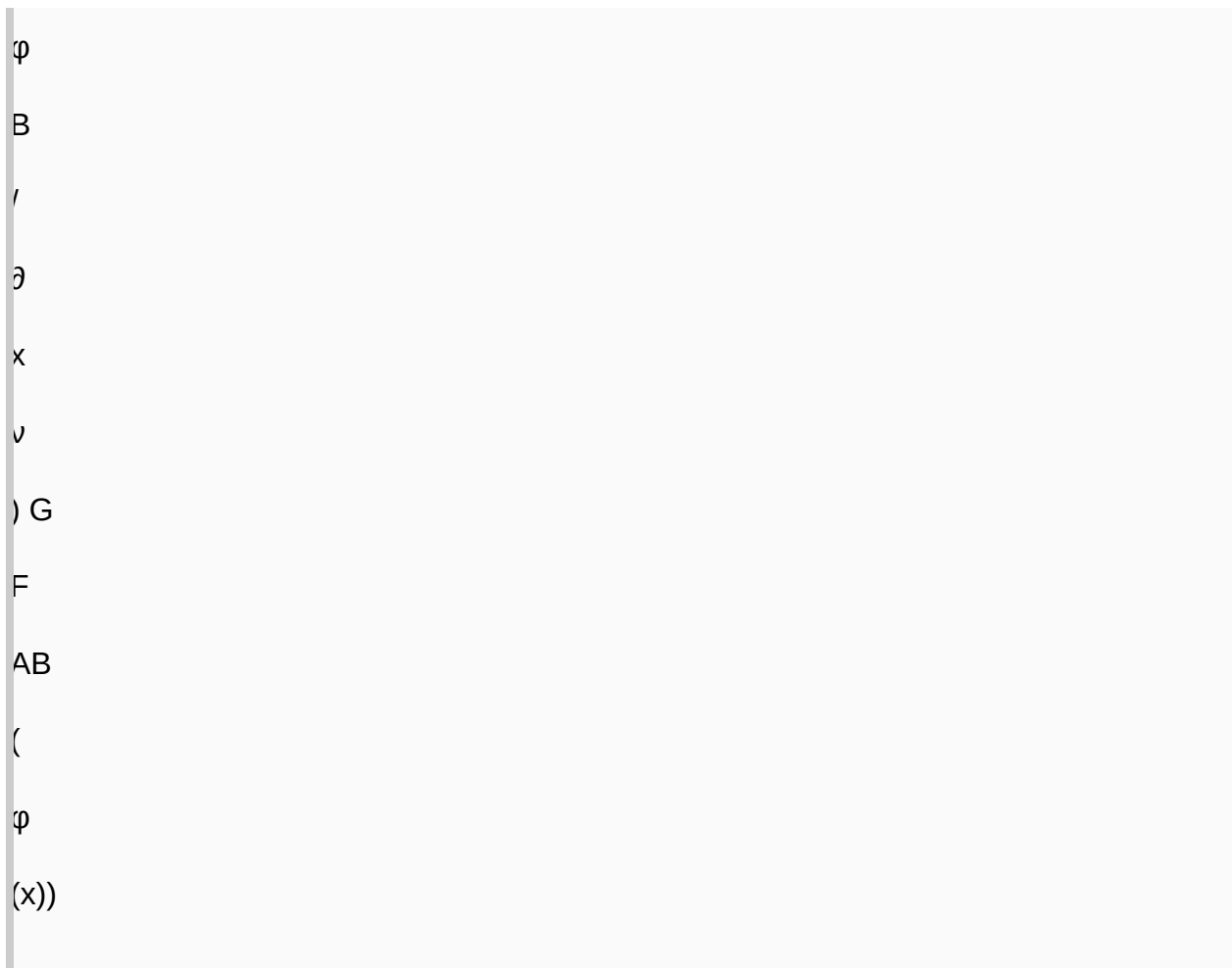
φ

$(x))$

For ToE: suppose the information manifold $\mathcal{A}$ maps to spacetime M via an emergence map $\varphi: \mathcal{A} \rightarrow M$. Then the spacetime metric is the pullback of the Fisher-Rao metric on the information manifold:

(11.12)

$$g_{\mu\nu}(x) = (\varphi^* G)_{\mu\nu} = \left(\frac{\partial \varphi^a}{\partial x^\mu} \frac{\partial \varphi^b}{\partial x^\nu} \right) G_{ab}(\varphi(x))$$



This formula — the induced metric equation — is how the geometry of the information manifold "flows down" to the geometry of spacetime. It is the ToE version of the Nash embedding theorem: the spacetime metric is induced by the information metric through the emergence map. The precise form of the emergence map and the conditions under which the induced metric is Lorentzian (rather than Riemannian) are determined by the Obidi Transformation, as will be shown in Part II.

Scholium 11.1 — On Coarse-Graining as Geometry

The concept of "coarse-graining" — integrating out microscopic degrees of freedom to obtain an effective macroscopic description — is

ubiquitous in physics. In the renormalization group, coarse-graining in momentum space transforms a high-energy theory to a low-energy effective theory. In kinetic theory, coarse-graining in momentum space (the fiber integral, equation 11.9) transforms the microscopic phase-space description to the macroscopic hydrodynamic description. In ToE, coarse-graining in statistical fiber space (equation 11.10) transforms the microscopic information-geometric description to the macroscopic spacetime description. The mathematical language of fiber integrals makes all these coarse-graining operations instances of the same geometric operation: the pushforward along a fiber bundle projection. This unification of coarse-graining operations under the single heading of "fiber integration" is one of the conceptual achievements of the fiber bundle language, and it makes ToE's emergence mechanism mathematically precise rather than merely metaphorical.



Section 12. Connections, Covariant Derivatives, and Parallel Transport

12.1 The Problem with Ordinary Derivatives on Curved Manifolds

On a curved manifold, the ordinary partial derivative $\partial_\mu V^\nu$ of a vector field V^ν does not transform as a tensor. To see why: under a coordinate change $x^\mu \rightarrow x'^\mu$, a vector field transforms as $V'^\mu = (\partial x'^\mu / \partial x^\nu) V^\nu$. Computing the partial derivative of this:

(12.1)

∂

V'

μ

γ

∂

x'

ρ

$= ($

∂

x'

μ

γ

∂

x

v

$)$

∂

V

v

γ

∂

x

σ

(

∂

x

σ

/

∂

x'

ρ

) + (

∂

2

x'

μ

/

∂

x

ν

∂

x

σ

) V
 ν
 (
 ∂
 x
 σ
 $/$
 ∂
 x'
 ρ
)

The first term is the expected tensor transformation law. The second term, involving the second derivative of the coordinate transformation, is an unwanted additional piece that prevents $\partial_\mu V^\nu$ from being a tensor. On a flat manifold with Cartesian coordinates, coordinate changes are linear and the second derivatives vanish; the problem does not arise. On a curved manifold, or in curvilinear coordinates, it does.

The solution is to define a covariant derivative that transforms tensorially by adding a "correction term" that exactly cancels the problematic second-derivative piece.

12.2 Affine Connections and the Covariant Derivative

Definition 12.1 (Affine Connection). An *affine connection* (or linear connection) on M is a rule ∇ that assigns to each pair (vector field X , vector field V) a vector field $\nabla_X V$, called the covariant derivative of V in the direction X , satisfying:

- Linearity in X : $\nabla_{X+Y} V = \nabla_X V + \nabla_Y V$, $\nabla_{fX} V = f \nabla_X V$.
- Linearity in V : $\nabla_X (V+W) = \nabla_X V + \nabla_X W$.
- Leibniz rule: $\nabla_X (fV) = X(f)V + f \nabla_X V$.

In local coordinates, the covariant derivative of a vector field $V = V^\nu \partial/\partial x^\nu$ in the coordinate direction $\partial/\partial x^\mu$ is:

(12.2)

$$\nabla_{\frac{\partial}{\partial x^\mu}} V = \frac{\partial V^\nu}{\partial x^\mu} \frac{\partial}{\partial x^\nu} + V^\nu \nabla_{\frac{\partial}{\partial x^\mu}} \frac{\partial}{\partial x^\nu}$$

$$\Gamma^{\rho}_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} \left(\nabla_{\mu} g_{\sigma\nu} + \nabla_{\nu} g_{\sigma\mu} - \nabla_{\sigma} g_{\mu\nu} \right)$$

where the $\Gamma^{\rho}_{\mu\nu}$ are the *connection coefficients* (Christoffel symbols). The term $\Gamma^{\rho}_{\mu\nu} V^{\nu}$ is the correction term that cancels the problematic second-derivative piece in the transformation law. The quantity $\nabla_{\mu} V^{\nu}$ is a (1,1) tensor — it has one upper index (from V) and one lower index (from the direction of differentiation) and transforms tensorially.

12.3 The Levi-Civita Connection

On a Riemannian or pseudo-Riemannian manifold, there is a preferred connection — the *Levi-Civita connection* — defined by two requirements:

- **Torsion-free:** $\Gamma^{\rho}_{\mu\nu} = \Gamma^{\rho}_{\nu\mu}$ (symmetric in lower indices).
- **Metric compatibility:** $\nabla_{\rho} g_{\mu\nu} = 0$ (the metric is covariantly constant).

These two conditions together uniquely determine the connection coefficients:

(12.3)

$$\Gamma^{\rho}_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} \left(\nabla_{\mu} g_{\sigma\nu} + \nabla_{\nu} g_{\sigma\mu} - \nabla_{\sigma} g_{\mu\nu} \right)$$

$\mu\nu$

$=$

$\frac{1}{2}$

g

$\rho\sigma$

$($

∂

μ

g

$\nu\sigma$

$+$

∂

ν

g

$\mu\sigma$

$-$

∂

σ

g

$\mu\nu$

$)$

This is the explicit formula for the Christoffel symbols of the Levi-Civita connection. Every component of the connection is determined by the metric and its first derivatives. The connection is thus not an independent geometric structure; it is determined by the metric.

Metric compatibility $\nabla_\rho g_{\mu\nu} = 0$ means that the inner product of vectors is preserved under parallel transport (which we define below). This is the geometric expression of the physical requirement that norms and angles are preserved under parallel translation along geodesics — a necessary condition for a consistent notion of "straight lines" on a curved manifold.

12.4 Covariant Derivative on Tensor Fields

The covariant derivative extends naturally to tensor fields of any type. For a covector field ω_ν :

(12.4)

$$\nabla_\mu \omega_\nu = \partial_\mu \omega_\nu - \Gamma_{\mu\nu}^\lambda \omega_\lambda$$

$$\begin{aligned}
 & - \\
 & \Gamma \\
 & \rho \\
 & \mu\nu \\
 & \omega \\
 & \rho
 \end{aligned}$$

Note the minus sign (opposite to the vector case). For a (0,2) tensor $T_{\mu\nu}$:

(12.5)

$$\begin{aligned}
 & \nabla \\
 & \rho \\
 & T \\
 & \mu\nu \\
 & = \\
 & \partial \\
 & \rho \\
 & T \\
 & \mu\nu \\
 & - \\
 & \Gamma \\
 & \sigma
 \end{aligned}$$

ρ_μ

Γ

σ_ν

$-$

Γ

σ

ρ_ν

Γ

$\mu\sigma$

The general rule: for each upper index, add a $+\Gamma$ term; for each lower index, subtract a Γ term. This ensures that the covariant derivative of any tensor is itself a tensor.

The covariant derivative of the entropy gradient (a covector) in ToE is:

(12.6)

∇

ρ

$($

∂

ν

$S) =$

∂
ρ
∂
ν
S
$-$
Γ
σ
$\rho\nu$
∂
σ
S

This symmetric (0,2) tensor, the Hessian of the entropy field, plays a central role in the Master Entropic Equation. Its trace $g^{\rho\nu}\nabla_\rho\partial_\nu S = \Box S$ is the d'Alembertian (wave operator) applied to S , which appears in the MEE as the kinetic term for the entropy field.

12.5 Parallel Transport

Definition 12.2 (Parallel Transport). A vector field V^μ is *parallel-transported* along a curve $\gamma(t)$ if:

(12.7)

DV

$$\begin{aligned}
 & \mu \\
 & \frac{d}{dt} = \left(\frac{dx^\mu}{dt} \right) \\
 & \frac{d}{dt} \\
 & \nabla \\
 & \frac{d}{dt} \\
 & \frac{d}{dt} \\
 & \mu \\
 & = dV \\
 & \mu \\
 & \frac{d}{dt} + \\
 & \Gamma \\
 & \mu \\
 & \nu \rho \\
 & \left(\frac{dx^\mu}{dt} \right) \\
 & \frac{d}{dt} V \\
 & \rho \\
 & = 0
 \end{aligned}$$

This is a first-order linear ODE along γ . Given an initial vector $V^\mu(\gamma(0))$, equation (12.7) determines uniquely the "parallel-transported" vector $V^\mu(\gamma(t))$

for all t . Parallel transport preserves the inner product of vectors when the connection is metric-compatible: $d/dt [g_{\mu\nu} V^\mu W^\nu] = 0$ whenever both V and W are parallel-transported along the same curve.

12.6 Geodesics

Definition 12.3 (Geodesic). A *geodesic* is a curve $\gamma(t)$ whose tangent vector dx^μ/dt is parallel-transported along itself:

(12.8)

$$d^2x^\mu/dt^2 + \Gamma^\mu_{\nu\rho} (dx^\nu/dt) (dx^\rho/dt) = 0$$

$$\frac{d}{dt} = 0$$

This is the *geodesic equation*. In GR, freely falling particles (not subject to any non-gravitational forces) follow geodesics of the spacetime metric. The geodesic equation is the GR generalization of Newton's first law: a particle moves in a "straight line" (geodesic) unless acted on by a force.

In ToE, entropic geodesics are the paths along which entropic configurations propagate without dissipation. The entropic geodesic equation is obtained from the Obidi Metric's connection, and in the near-equilibrium limit, it reduces to the geodesic equation (12.8) of GR with the effective spacetime metric given by the Obidi Metric.

12.7 The Amari-Čencov α -Connections

On a statistical manifold, the Levi-Civita connection of the Fisher-Rao metric is only one member of a family of connections — the Amari-Čencov α -connections.

Definition 12.4 (α -Connection). For any $\alpha \in \mathbb{R}$, the α -connection $\nabla^{(\alpha)}$ on a statistical manifold has connection coefficients:

(12.9)

$$\Gamma_{ijk}^{(\alpha)} =$$

Γ (0) $_{ijk}$ $-$ $($ α $/2) C$ $_{ijk}$

where $\Gamma^{(0)}$ are the Christoffel symbols of the Fisher-Rao metric (the $\alpha = 0$ Levi-Civita case) and C_{ijk} is the Amari-Čencov cubic tensor (skewness tensor) of the distribution family:

(12.10)

 C $_{ijk}$ $($ θ $) =$ $[$ $p(x|$ θ

) (

∂

∂

$\log p) ($

∂

∂

$\log p) ($

∂

k

$\log p) dx$

This is the third central moment of the score function (the log-likelihood gradient). For $\alpha = 0$: the Levi-Civita (metric-compatible) connection. For $\alpha = +1$: the exponential connection $\nabla^{(e)}$, natural for exponential families. For $\alpha = -1$: the mixture connection $\nabla^{(m)}$, natural for mixture families.

The remarkable duality: the α and $-\alpha$ connections are dual with respect to the Fisher-Rao metric in the sense that:

(12.11)

$X \cdot G$

F

$(Y, Z) = G$

F
 $($
 ∇
 (α)
 X
 $Y, Z) + G$
 F
 $(Y,$
 ∇
 $(-\alpha)$
 X
 $Z)$

This duality structure — a Riemannian metric with two dual connections — is the hallmark of the "dually flat" geometry that Amari and Nagaoka identified as the appropriate setting for exponential families. In ToE, the classical sector of the Hybrid Metric-Affine Space uses the $\alpha = 0$ connection (equilibrium, reversible dynamics), while the non-equilibrium, irreversible sector uses $\alpha \neq 0$ connections. The transition between equilibrium and non-equilibrium regimes in ToE corresponds to a transition between the $\alpha = 0$ and $\alpha \neq 0$ connection, governed by the Obidi Curvature Invariant (Section 13).

Scholium 12.1 — The Physical Meaning of the α -Connections in ToE

The family of α -connections encodes, in a single mathematical object, the spectrum of dynamical regimes from perfectly reversible equilibrium to maximally irreversible non-equilibrium. At $\alpha = 0$ (Levi-Civita connection), the parallel transport is metric-compatible: norms are preserved, processes are reversible, and the system is at thermodynamic equilibrium. As $|\alpha|$ increases, the connection departs from metric compatibility in a controlled way that encodes increasing irreversibility. The Amari-Čencov cubic tensor C_{ijk} , which measures the departure from Gaussianity (the third moment of the score function), controls the deviation from the $\alpha = 0$ case. In ToE, this has a direct physical interpretation: a Gaussian entropic distribution (characterized by vanishing C_{ijk}) is a thermodynamic equilibrium configuration. Deviations from Gaussianity (non-zero C_{ijk} , $\alpha \neq 0$) correspond to irreversible entropic processes. The Master Entropic Equation, derived from the Obidi Action, determines how the entropy field evolves and, in so doing, determines the appropriate α at each point of the manifold. This is a remarkable unification: a single mathematical object (α -connections) encodes the full thermodynamic spectrum from reversible equilibrium to irreversible non-equilibrium.



Section 13. The Curvature of a Manifold

13.1 The Riemann Curvature Tensor

Curvature is the phenomenon by which parallel transport around a closed loop fails to return a vector to its original direction. On a flat manifold, parallel transport is path-independent; on a curved manifold, it depends on the path. The Riemann curvature tensor measures this path-dependence.

Definition 13.1 (Riemann Curvature Tensor). The *Riemann curvature tensor* R of an affine connection ∇ is defined by:

(13.1)

$$R(X,Y)Z =$$

$$\nabla_X$$

$$\nabla_Y$$

$$\nabla_X$$

$$\nabla_Y$$

$$\nabla_X$$

$$\nabla_Y$$

$$\nabla_X$$

$$\nabla_Y$$

$$\nabla_X$$

$$\nabla_Y$$

Z

$-$

∇

$[X, Y]$

Z

where $[X, Y]$ is the Lie bracket of vector fields. In component notation, with the shorthand $R(X, Y)Z = R(\partial_\mu, \partial_\nu)V^\rho Z = R^\rho_{\sigma\mu\nu}V^\sigma$:

(13.2)

R

ρ

$\sigma\mu\nu$

$=$

∂

μ

Γ

ρ

$\nu\sigma$

$-$

∂

ν

$$\begin{aligned}
 & \Gamma_{\rho\mu\sigma} \\
 & + \\
 & \Gamma_{\rho\mu\lambda} \\
 & \Gamma_{\lambda\nu\sigma} \\
 & - \\
 & \Gamma_{\rho\nu\lambda} \\
 & \Gamma_{\lambda\mu\sigma}
 \end{aligned}$$

This is the fundamental formula for the Riemann tensor. It is a type (1,3) tensor — one upper index and three lower indices. The commutator interpretation (13.1) captures the path-dependence: if you parallel-transport a vector first in direction X then Y , versus first Y then X , the results differ by $R(X,Y)Z$. Nonzero curvature means nonzero difference — the manifold is curved.

13.2 Symmetries of the Riemann Tensor

With the metric, all indices can be lowered to give $R_{\mu\nu\rho\sigma} = g_{\mu\lambda} R^{\lambda}_{\nu\rho\sigma}$. The Riemann tensor then satisfies the following symmetry identities:

(13.3)

$$R_{\mu\nu\rho\sigma} = -R_{\nu\mu\rho\sigma} \\ = -R_{\mu\nu\sigma\rho} \\ = R_{\rho\sigma\mu\nu} \\ \text{(antisymmetric in first pair)}$$

(13.4)

$$R_{\mu\nu\rho\sigma} = -R_{\mu\nu\sigma\rho} \\ = R_{\rho\sigma\mu\nu} \\ \text{(antisymmetric in second pair)}$$

(13.5)

$$R_{\mu\nu\rho\sigma} = R_{\rho\sigma\mu\nu}$$

(symmetric under pair exchange)

(13.6)

$$R_{\mu}{}^{\nu}{}_{[\rho\sigma]} = 0$$

(first Bianchi identity)

where brackets denote antisymmetrization over the indicated indices. These symmetries reduce the number of independent components of the Riemann tensor in 4 dimensions from $4^4 = 256$ to 20.

13.3 The Ricci Tensor and Ricci Scalar

Definition 13.2 (Ricci Tensor). The *Ricci tensor* is the trace of the Riemann tensor on the first and third indices:

(13.7)

$$R_{\mu\nu} = R_{\nu\mu}$$

$$R_{\mu\nu} = g_{\mu\rho} R^{\rho}_{\nu}$$

$$R = R^{\mu}_{\mu}$$

$$R_{\mu\nu} = R_{\nu\mu}$$

The Ricci tensor is symmetric: $R_{\mu\nu} = R_{\nu\mu}$. It has 10 independent components in 4 dimensions. It measures the average curvature in planes containing each direction: $R_{\mu\mu}$ (no sum) is the average of sectional curvatures in planes containing the μ -direction.

Definition 13.3 (Ricci Scalar). The *Ricci scalar* (or scalar curvature) is the trace of the Ricci tensor:

(13.8)

$$R = g^{\mu\nu} R_{\mu\nu}$$

This is a scalar — a single number at each point that measures the total curvature of the manifold at that point. In 2 dimensions, R is twice the Gaussian curvature. For a round 2-sphere of radius r , $R = 2/r^2$. For flat Euclidean space, $R = 0$.

13.4 The Einstein Tensor

Definition 13.4 (Einstein Tensor). The *Einstein tensor* is:

(13.9)

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$$

The Einstein tensor is symmetric. Its crucial property is that it is automatically divergence-free:

(13.10)

$$\nabla_\mu G^{\mu\nu} = 0$$

$$\nabla_\mu G^{\mu\nu} = 0$$

This is a purely geometric identity that holds for any Riemannian or pseudo-Riemannian manifold. It is derived from the contracted Bianchi identity (Section 13.5) and does not require the Einstein field equations. It is a consequence of the geometry alone.

The significance of (13.10) for physics is profound: the right-hand side of the Einstein field equations, $T_{\mu\nu}$, must also be divergence-free: $\nabla^\mu T_{\mu\nu} = 0$. This is the local conservation law for energy and momentum. The Einstein tensor is the unique divergence-free, symmetric, second-rank tensor expressible in terms of the metric and its first and second derivatives. This uniqueness, proved by David Lovelock in 1971, means that if one wants a local, generally covariant, second-order field equation for the metric sourced by matter, the Einstein field equations with a cosmological constant are essentially the only option in 4 dimensions.

13.5 The Bianchi Identity

Theorem 13.1 (Bianchi Identity). The Riemann tensor satisfies the *second Bianchi identity*:

$$(13.11)$$

∇

$[\lambda$

R

$\mu\nu]\rho\sigma$

$= 0$

,

equivalently:

∇

λ

R

$\mu\nu\rho\sigma$

$+$

∇

μ

R

$\nu\lambda\rho\sigma$

$+$

∇

ν

R

$\lambda\mu\rho\sigma$

$= 0$

Contracting (13.11) with $g^{\mu\rho}$:

(13.12)

$$\begin{aligned} & \nabla_\mu \lambda \\ & R_{\mu\sigma} \\ & - \\ & \nabla_\mu \nu \\ & R_{\mu\sigma} \\ & \lambda\sigma \\ & + \\ & \nabla_\mu \\ & \mu \\ & R_{\mu\sigma} \\ & \mu\lambda\nu\sigma \\ & = 0 \end{aligned}$$

Contracting again with $g^{\nu\sigma}$:

(13.13)

$$\nabla_\mu$$

λ

R

$-$

2

∇

ν

R

$\nu\lambda$

$= 0$

$,$

i.e.,

∇

ν

G

$\nu\lambda$

$=$

∇

ν

$(R$

$\nu\lambda$

$-$

$\frac{1}{2}$

g

$\nu\lambda$

$R) = 0$

This is the contracted Bianchi identity, which immediately gives the divergence-free property (13.10) of the Einstein tensor. Note that equation (13.10) is not a law; it is a theorem — a purely geometric consequence of the definition of the Riemann tensor via the Levi-Civita connection. It holds regardless of whether the metric satisfies the Einstein field equations.

13.6 Physical Interpretation of Curvature

In GR, the Einstein field equations are $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$. The contracted Bianchi identity (13.10) then implies $\nabla^\mu T_{\mu\nu} = 0$ — the covariant conservation of energy and momentum. This is how conservation laws are encoded in curved spacetime: not as $\partial_\mu T^{\mu\nu} = 0$ (which would not be covariant), but as $\nabla_\mu T^{\mu\nu} = 0$ (which is covariant and includes the effects of curved spacetime on the energy-momentum flow).

In ToE, the analogous conservation law is $\nabla^\mu T^{\text{ent}}_{\mu\nu} = 0$, which emerges from the contracted Bianchi identity of the Obidi Metric. This ensures that the entropic energy-momentum is conserved under the entropic field dynamics, just as matter energy-momentum is conserved under the GR field equations.

13.7 The Obidi Curvature Invariant (OCI)

The Obidi Curvature Invariant (OCI) is the most fundamental curvature quantity in the Theory of Entropicity. It is not the Ricci scalar of GR; it is a

deeper, information-geometric curvature invariant that sets the quantum of informational curvature on the statistical manifold.

Definition 13.5 (Obidi Curvature Invariant). The *Obidi Curvature Invariant* (OCI) is the scalar:

(13.14)

$$\text{OCI} = \ln 2$$

This is not an approximation or a numerical coincidence. It is the exact, dimensionless, information-geometric curvature invariant of the entropic manifold, established by Obidi as the minimum distinguishability curvature of a statistical manifold with a binary choice at its base.

Why does $\ln 2$ appear? Because $\ln 2$ is the Shannon entropy of a perfectly balanced binary random variable (a fair coin): a binary random variable with outcomes $\{0, 1\}$ each with probability $1/2$ has entropy $H = -\frac{1}{2} \ln \frac{1}{2} - \frac{1}{2} \ln \frac{1}{2} = \ln 2$. This is the informational measure of "one bit" — the minimum amount of information required to distinguish between two equally likely alternatives. It is the quantum of informational curvature: the minimum curvature that a statistical manifold must possess in order for adjacent points to be statistically distinguishable.

The OCI enters the ToE formalism in several ways:

- It sets the scale of the Obidi Metric: the Obidi Transformation maps the Fisher-Rao metric to the Obidi Metric by a rescaling whose fundamental constant is $\text{OCI} = \ln 2$.

- It determines the quantum of mass in ToE: the minimum mass of a stable entropic configuration is proportional to $OCI / (\text{Vuli-Ndlela Integral})$, a ratio that will be computed explicitly in Part II.
- It separates the classical sector (entropy fluctuations with curvature $> OCI$) from the quantum sector (entropy fluctuations with curvature $< OCI$) of the Hybrid Metric-Affine Space. The HMAS boundary is the hypersurface of constant curvature $K = OCI = \ln 2$.
- It explains why Planck's constant $\hbar$ is not zero: the irreducible binary distinguishability ($OCI = \ln 2$) prevents quantum systems from having classical (zero-variance) probability distributions. The quantum of action is the entropic consequence of the impossibility of perfect binary discrimination without a cost of at least $k_B \ln 2$.

Scholium 13.1 — On the Profound Simplicity of the $OCI = \ln 2$

The discovery that the fundamental curvature invariant of the entropic manifold is $\ln 2$ deserves to be properly appreciated. In all of physics, the fundamental constants — Newton's G , Planck's $\hbar$, the speed of light c , Boltzmann's k_B — are empirical parameters whose values must be measured. They have no theoretical derivation within the current framework; they are given by experiment and entered by hand. The $OCI = \ln 2$, by contrast, is a mathematical truth: it is the information entropy of a fair binary choice, which is derivable from the axioms of information theory alone. Its appearance as the fundamental curvature invariant of the entropic manifold suggests that the universe's deepest constants may not be empirical parameters at all, but mathematical necessities — theorems that follow from the logical structure of information and distinguishability. If this is correct, it represents one of the most

significant conceptual achievements in the history of physics: the replacement of empirical constants with mathematical theorems. The Theory of Entropicity places this remarkable discovery at the center of its framework, and its implications will be fully worked out in Parts II and III.



Section 14. Moments of Distributions — The Complete Theory

14.1 What Are Moments?

For a probability distribution $p(x)$ on $\mathbb{R}^n$, the k -th moment tensor is the expectation value of the symmetric product of k coordinate functions:

(14.1)

M
μ
1
...
μ
k

$$\begin{aligned}
 &= \langle x^\mu \rangle \\
 &= \int x^\mu p(x) dx \\
 &= \int x^\mu p(x) dx
 \end{aligned}$$

The moments encode all statistical properties of the distribution: the zeroth moment is normalization, the first moment is the mean, the second moment

encodes variance and correlations, and higher moments encode increasingly refined geometric and statistical properties. In the context of ToE, moments of the local entropic distribution function are the bridge between the microscopic information-geometric picture and the macroscopic tensor structure of general relativity.

14.2 The Zeroth Moment: Normalization

The zeroth moment is simply:

(14.2)

$$M^{(0)} = \int p(x) dx = 1$$

This is the normalization condition. It says the total probability is 1, or equivalently that the total entropic weight of all microstates is conserved. In ToE, this condition translates to the conservation of total entropic "charge" — the integrated entropy density over any Cauchy surface is conserved under the dynamics of the MEE in the absence of sources. This is the zeroth conservation law of ToE, analogous to the conservation of total electric charge in electrodynamics.

14.3 The First Moment: Mean and Energy-Momentum Flux

The first moment (mean) of a distribution $p(x)$ is the vector:

(14.3)

$$\begin{aligned} \mathbf{M} \\ \mu \\ = \\ \langle \\ x \\ \mu \\ \rangle \\ = \\ \int \\ x \\ \mu \\ p(x) \, d \\ n \\ x \end{aligned}$$

This is the center of mass of the distribution — the "most probable position" in an average sense. In a kinetic theory context, if p is the momentum distribution, the first moment gives the average momentum:

(14.4)

which is the particle number current (or energy flux, depending on the normalization). In ToE, the first moment of the entropic momentum distribution:

(14.5)

x
q
μ
f
ent
(x, q) d
μ
F
(q)

where $q^\mu = G^{F,\mu\nu} \partial_\nu S$ is the contravariant entropy gradient (entropic momentum), gives the entropic particle number current — the net flow of entropy per unit spacetime volume. This is identified in ToE with the matter number current of classical field theory in the near-equilibrium limit: $J^\mu_{ent} \rightarrow n u^\mu$ where n is the matter number density and u^μ is the four-velocity.

14.4 The Second Moment: Covariance and the Stress-Energy Tensor

The second moment tensor of a distribution $p(x)$ is:

(14.6)

M
$\mu\nu$

$$\begin{aligned}
 &= \\
 &\langle \\
 &x \\
 &\mu \\
 &x \\
 &\nu \\
 &\rangle \\
 &= \\
 &\int \\
 &x \\
 &\mu \\
 &x \\
 &\nu \\
 &p(x) \, d \\
 &n \\
 &x
 \end{aligned}$$

The covariance tensor, which measures the spread and correlations of the distribution, is:

(14.7)

$$\Sigma$$

$\mu\nu$ $=$ $\langle$ x μ x ν $\rangle$ $-$ $\langle$ x μ $\rangle$ $\langle$ x ν $\rangle$

This is a rank-2 symmetric tensor. It is positive semi-definite for any probability distribution (by the Cauchy-Schwarz inequality). In kinetic theory, the second moment of the momentum distribution gives the stress (pressure) tensor. Explicitly, for a relativistic gas:

(14.8)

T
 $\mu\nu$
 $(x) =$
 $\int$
 p
 μ
 p
 v
 $f(x, p) ($
 $-$
 $g)$
 $1/2$
 d
 3
 p / p
 0

where the integral runs over the mass shell $g_{\mu\nu}p^\mu p^\nu = -m^2 c^2$. The energy density is T^{00} , the momentum density is T^{0i}/c , and the pressure tensor is T^{ij} . The trace of the spatial part gives the thermodynamic pressure $P = T^{ii}/3$. For a perfect fluid in its rest frame, $T^{\mu\nu} = \text{diag}(\rho c^2, P, P, P)$.

The deepest insight: **the stress-energy tensor is the second moment of the momentum distribution.** This is not an approximation; it is the exact

definition in kinetic theory. In ToE, the Entropic Stress-Energy Tensor is defined analogously as the second moment of the entropic momentum distribution:

(14.9)

$$T^{\mu\nu}_{ent}(x) = \int \wedge x q \mu q v f_{ent}(x, q) d\mu F(q)$$

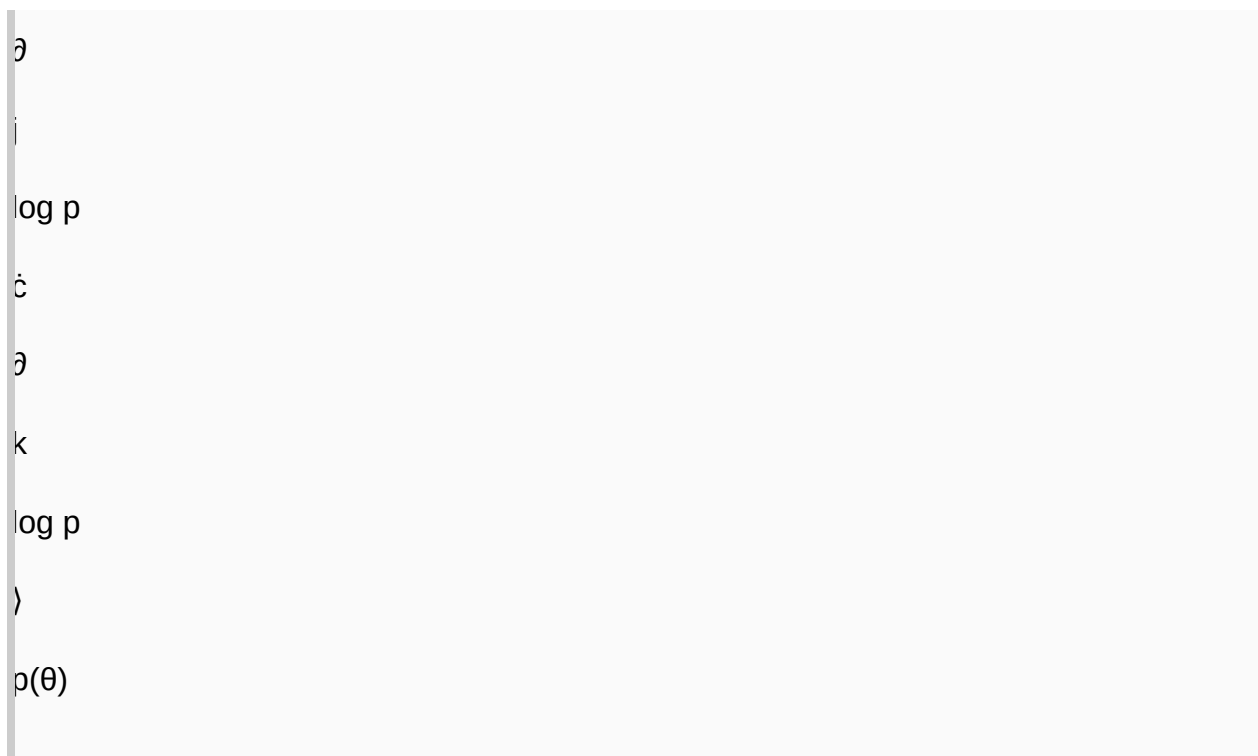
where $q_\mu = \partial_\mu S$ is the entropic momentum covector and $f^{ent}(x, q)$ is the entropic distribution function derived from the Obidi Action. This is a fiber integral (Section 11.6) over the statistical fiber Λ_x at each spacetime point x . The Entropic Stress-Energy Tensor so obtained is symmetric (since $q_\mu q_\nu = q_\nu q_\mu$), satisfies $\nabla^\mu T^{ent}_{\mu\nu} = 0$ by the Bianchi identity, and in the near-equilibrium limit reduces to $(c^4/8\pi G)G_{\mu\nu}$ — recovering the Einstein field equations.

14.5 The Third Moment: Skewness and the Amari-Čencov Tensor

The third central moment measures the asymmetry (skewness) of a distribution. On a statistical manifold, the fundamental third-order geometric object is the Amari-Čencov cubic tensor:

(14.10)

C
ijk
(
θ
) =
(
θ
i
log p
ċ



where the expectation is taken with respect to $p(x/\theta)$. This is the third moment of the score function $\partial_i \log p$ — the gradient of the log-likelihood. The cubic tensor is a completely symmetric (0,3) tensor on the statistical manifold. It vanishes for Gaussian distributions (which are symmetric and have zero skewness) and is non-zero for any non-Gaussian distribution.

In information geometry, the cubic tensor is the "third derivative" of the entropy functional and controls the deviation from the dually flat geometry of exponential families. In ToE, it controls the magnitude of the α -connection departure from the Levi-Civita connection (as in equation 12.9). A distribution with large C_{ijk} is highly non-Gaussian, highly asymmetric, and corresponds in the ToE picture to a strongly irreversible entropic configuration. Such configurations drive rapid entropy production and correspond to the far-from-equilibrium regime of the Master Entropic Equation.

14.6 The Cramér-Rao Bound and Fundamental Limits

The moments of a distribution are constrained by fundamental inequalities. The most important is the *Cramér-Rao bound*, which says that the variance of any unbiased estimator $\hat{\theta}$ of the parameter θ satisfies:

(14.11)

$$\text{Var}(\hat{\theta}) \geq 1 / I(\theta)$$

where $I(\theta) = \langle (\partial_{\theta} \log p)^2 \rangle = G^F_{\theta\theta}$ is the Fisher information. In the multivariate case:

(14.12)

$$\text{Cov}(\hat{\theta})$$

$\geq$ $(G$ F $)$ -1

(matrix inequality)

The Cramér-Rao bound says: the precision of any statistical measurement is limited by the Fisher information. The Fisher-Rao metric measures how quickly the distribution changes as the parameter changes; high Fisher information means the distribution is sensitive to small parameter changes, which means the parameter can be estimated precisely. Low Fisher information means insensitivity, hence poor estimation.

In ToE, the Cramér-Rao bound has a direct physical interpretation: it sets a fundamental limit on the precision with which the entropy field $S(x)$ can be localized. In the ToE picture, the parameter θ corresponds to the configuration of the entropy field, and the bound (14.12) says that entropy configurations with low Fisher information (low distinguishability) cannot be precisely localized. This is the ToE analog of Heisenberg's uncertainty principle: just as quantum mechanics limits the simultaneous precision of position and momentum by $\Delta x \Delta p \geq \hbar/2$, ToE limits the precision of entropy field localization by the Cramér-Rao bound on the statistical manifold. In both cases, the fundamental limit arises from the irreducible informational structure of the theory — Heisenberg's from quantum probability amplitudes, ToE's from classical (or quantum) probability distributions on the statistical

manifold. The $\text{OCI} = \ln 2$ sets the scale of this fundamental indistinguishability.

14.7 The Fiber Integral Connection: From Distributions to Tensors

The synthesis of all the ideas in this section can now be stated in full generality. The passage from microscopic probability distributions to macroscopic tensors — specifically, to the stress-energy tensor and the Einstein tensor — is accomplished by the sequence of operations:

4. **Specify the microscopic distribution:** At each spacetime point x , specify the local probability distribution $f^{ent}(x, q)$ over entropic momentum configurations $q_\mu = \partial_\mu S$. This is a section of a bundle over M whose fiber is the space of distributions on the entropic momentum space.
5. **Compute the second moment (fiber integral):** Integrate $q_\mu q_\nu f^{ent}$ over the statistical fiber at each x , using the Fisher-Rao volume form $d\mu_F$. This gives the Entropic Stress-Energy Tensor $T^{ent}_{\mu\nu}(x)$ as a function on the base manifold M . The operation is equation (14.9).
6. **Apply the Obidi Transformation:** Convert the Riemannian information metric to the Lorentzian Obidi Metric. This establishes the spacetime geometry.
7. **Compute the Einstein tensor:** From the Obidi Metric, compute the Christoffel symbols (12.3), the Riemann tensor (13.2), the Ricci tensor (13.7), the Ricci scalar (13.8), and the Einstein tensor (13.9).
8. **Identify the near-equilibrium limit:** Show that in the limit of small entropy gradients and near-equilibrium distributions (nearly Gaussian, $C_{ijk} \approx 0$), the equation $G_{\mu\nu} = (8\pi G/c^4) T^{ent}_{\mu\nu}$ holds as a

theorem, not an axiom. This is the complete derivation of the Einstein field equations from the Theory of Entropicity.

Steps 1 through 5 are what Parts II and III of this Letter will carry out in full mathematical detail. Part I has now established all the mathematical tools needed for that derivation: manifolds (Section 5), tangent and cotangent bundles (Sections 6 and 7), tensors (Section 8), fiber bundles and fiber integrals (Sections 9 and 11), probability distributions and statistical manifolds (Section 10), covariant derivatives and connections (Section 12), curvature (Section 13), and moments (this section). The stage is set.

Scholium 14.1 — On the Unification of Kinetic Theory and Information Geometry

The fiber integral formula (14.9) for the Entropic Stress-Energy Tensor is structurally identical to the kinetic theory formula (14.8) for the ordinary stress-energy tensor, with one crucial difference: in kinetic theory, the distribution function $f(x,p)$ is an empirical quantity describing the actual momentum distribution of a gas of particles, and the stress-energy tensor is computed from it. In ToE, the distribution function $f^{ent}(x,q)$ is derived from the Obidi Action by an extremal principle — it is the most probable (maximum entropy) distribution consistent with the macroscopic constraints encoded in the entropy field $S(x)$. In other words, ToE derives the distribution function from first principles, whereas kinetic theory takes it as input. This is the precise sense in which ToE is more fundamental than kinetic theory: it provides the microscopic derivation of the distribution function from which kinetic theory then computes macroscopic quantities. The "particles" of kinetic theory are, in ToE, not primitives but emergent stable configurations (local minima

of the Obidi Action) of the entropy field. Their momentum distribution is not an independent input; it is determined by the dynamics of $S(x)$.



Section 15. Summary and Bridge to Part II

15.1 Recapitulation of the Revolutionary Inversion

Part I of this Letter has established, through five independent lines of argument, that the standard hierarchy of physics — in which matter is primitive and entropy is derived — must be inverted. The entropy field $S(x)$ is the ontological primitive. Matter, energy, radiation, and spacetime are all emergent from the dynamics of $S(x)$ as governed by the Obidi Action. This inversion is mandated by: (a) the Shannon-Landauer-Wheeler information primacy results; (b) the Bekenstein-Hawking-Jacobson-Verlinde thermodynamic gravity results; (c) Čencov's uniqueness theorem for the Fisher-Rao metric; (d) the standard non-equilibrium thermodynamic description of entropy as a local field; and (e) the logical economy argument. No single one of these arguments is conclusive alone; taken together, they constitute a compelling and, in the author's judgment, logically overwhelming case for the inversion.

15.2 Summary of Mathematical Tools Developed

Part I has developed the following mathematical framework, which constitutes the complete prerequisites for all subsequent work in Parts II and III:

Section	Mathematical Tool	Role in ToE
5	Topological spaces, differentiable manifolds, charts	Foundation for the entropy field $S(x)$ and the statistical manifold Λ
6	Tangent spaces, tangent bundle TM , vector fields	Entropic flow vector $(\nabla S)^\mu$; entropic geodesics
7	Cotangent spaces, cotangent bundle T^*M , one-forms	Entropy gradient $\partial_\mu S = dS$ as primary covector; Obidi Transformation input
8	Tensors, metric tensor, Einstein tensor, key tensors of ToE	Fisher-Rao metric, Obidi Metric, Entropic Stress-Energy Tensor $T^{\text{ent}}_{\mu\nu}$
9	Fiber bundles, vector bundles, principal bundles, connections	Statistical manifold bundle; HMAS structure; gauge formulation of ToE
10	Probability distributions, statistical manifolds, exponential families	Local distribution $p(y x)$; entropy field $S(x) = H(p_x)$
11	Differential forms, Stokes' theorem, fiber integrals, pullback	Coarse-graining; induced metric equation; Obidi Action integration
12	Connections, covariant derivatives, parallel transport, geodesics, α -connections	Levi-Civita connection of Obidi Metric; α -connections for irreversible dynamics

Section	Mathematical Tool	Role in ToE
13	Riemann tensor, Ricci tensor, Einstein tensor, Bianchi identity, OCI	LHS of Einstein field equations; OCI = $\ln 2$ as quantum of informational curvature
14	Moments of distributions, Cramér-Rao bound, fiber integral connection	$T^{\text{ent}}_{\mu\nu}$ as second moment; kinetic theory unification; RHS of EFE derivation

15.3 Bridge to Part II

Part II of this Letter will use all of the above to develop the complete information geometry of the Theory of Entropicity. Beginning from the Fisher-Rao metric (8.4), Part II will derive the complete geometric structure of the statistical manifold including the Fubini-Study metric for quantum statistical states, the full theory of Amari-Čencov α -connections and their duality, the Kullback-Leibler divergence and Araki-Umegaki quantum divergence, and the Rényi-Tsallis entropy generalizations relevant to non-extensive systems. Part II will then present the complete ToE core structure: the Obidi Action, derived from first principles as the most general action functional for a scalar field on a statistical manifold consistent with the OCI constraint; the Obidi Transformation, derived as the unique map converting the positive-definite Fisher-Rao metric to a Lorentzian metric by a controlled sign-flip of the temporal component; the Obidi Metric, the resulting Lorentzian metric on the information manifold; the Hybrid Metric-Affine Space (HMAS), the combined bundle structure with classical Fisher and quantum Fubini-Study sectors; the Master Entropic Equation (MEE), derived by variation of the Obidi Action; and the Vuli-Ndlela Integral, the specific fiber integral that extracts the macroscopic matter fields from the microscopic entropic dynamics.

15.4 The Fundamental Theorem

The fundamental theorem that this Letter IV, taken in its entirety (Parts I, II, and III), aims to prove is the following:

Fundamental Theorem of the Theory of Entropicity (ToE)

The Einstein field equations

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

are the low-entropy-gradient, near-equilibrium, long-wavelength limit of the entropic field equations derived from the Obidi Action of the Theory of Entropicity. Specifically: (i) the left-hand side $G_{\mu\nu}$ is the Einstein tensor of the Obidi Metric in the near-equilibrium limit; (ii) the right-hand side $T_{\mu\nu}$ is the near-equilibrium limit of the Entropic Stress-Energy Tensor $T^{\text{ent}}_{\mu\nu}$, which is the second moment (fiber integral) of the entropic momentum distribution derived from the Obidi Action; and (iii) the coupling constant $8\pi G/c^4$ is determined by the Obidi Curvature Invariant $OCI = \ln 2$ and the Vuli-Ndlela normalization of the fiber integral.

The complete proof of this theorem occupies Parts II and III of the present Letter. Part I has established all the mathematical language in which the proof will be conducted. The reader who has mastered Sections 5 through 14 possesses every mathematical tool required to follow the proof in its full detail.



Section 16. Table of Mathematical Symbols and Notation

The following table provides a comprehensive reference for all mathematical symbols used in Part I, organized by category. The Einstein summation convention (repeated upper-lower index pairs are summed) is used throughout unless explicitly stated otherwise.

16.1 Index Conventions

Index Type	Range	Usage
$\mu, \nu, \rho, \sigma, \lambda$	0, 1, 2, 3	Spacetime indices (4-dimensional)
i, j, k, l	1, 2, 3	Spatial indices (3-dimensional)
A, B, C, D	1, ..., n	Statistical manifold / information manifold indices
α, β	Chart label	Indices labeling coordinate charts in an atlas
a, b	1, ..., N	Particle labels in kinetic theory contexts

16.2 Manifolds and Topology

Symbol	Name	Type	Definition / ToE Role
M	Spacetime manifold	4-dim pseudo-Riemannian manifold	The emergent spacetime of ToE; base of the statistical bundle
Λ	Statistical (information) manifold	n-dim Riemannian manifold	Space of probability distributions; total space of the entropic bundle
$(U_\alpha, \varphi_\alpha)$	Coordinate chart	Open set + homeomorphism	Local coordinate system on M or Λ
$C^\infty(M)$	Smooth functions on M	Algebra of smooth maps $M \rightarrow \mathbb{R}$	$S(x) \in C^\infty(M)$ is the entropy field

Symbol	Name	Type	Definition / ToE Role
S^n	n-sphere	n-dim compact manifold	Model for statistical manifolds with spherical topology
Δ^{n-1}	(n-1)-simplex	Compact subset of $\mathbb{R}^n$	Space of discrete probability distributions on n outcomes

16.3 Tangent and Cotangent Structures

Symbol	Name	Type	Definition / ToE Role
$T_p M$	Tangent space at p	n-dim real vector space	Space of tangent vectors (directions) at point $p \in M$
TM	Tangent bundle	2n-dim manifold	Disjoint union of all tangent spaces; arena for vector fields
$\partial/\partial x^\mu _p$	Coordinate basis vector	Tangent vector (derivation)	Basis of $T_p M$ in coordinate chart (x^μ)
X^μ	Vector components	Array of n real numbers	Components of vector $X = X^\mu \partial/\partial x^\mu$
$T_p^* M$	Cotangent space at p	n-dim real vector space	Dual of $T_p M$; space of linear functionals on $T_p M$
$T^* M$	Cotangent bundle	2n-dim manifold	Disjoint union of all cotangent spaces; arena for one-forms

Symbol	Name	Type	Definition / ToE Role
dx^μ	Coordinate basis one-form	Covector	Dual basis satisfying $dx^\mu(\partial/\partial x^\nu) = \delta^\mu_\nu$
ω_μ	Covector components	Array of n real numbers	Components of one-form $\omega = \omega_\mu dx^\mu$
$\partial_\mu S$	Entropy gradient (covector)	Section of T^*M	Fundamental ToE covector; drives entropic forces
$(\nabla S)^\mu$	Entropy gradient (vector)	Section of TM	$g^{\mu\nu}\partial_\nu S$; requires Obidi Metric to define

16.4 Tensor Fields

Symbol	Name	Type	Definition / ToE Role
$g_{\mu\nu}$	Spacetime metric tensor	Symmetric (0,2) tensor, Lorentzian	Obidi Metric in the near-equilibrium limit; determines GR geometry
$g^{\mu\nu}$	Inverse metric tensor	Symmetric (2,0) tensor	$g^{\mu\rho}g_{\rho\nu} = \delta^\mu_\nu$
G^F_{AB}	Fisher-Rao metric	Symmetric (0,2) tensor on Λ , Riemannian	Canonical information metric on statistical manifold; Obidi Transformation input
$\tilde{G}_{AB}$	Obidi Metric	Symmetric (0,2) tensor, Lorentzian	Result of Obidi Transformation; identified with spacetime

Symbol	Name	Type	Definition / ToE Role
			metric
$T_{\mu\nu}$	Stress-energy tensor	Symmetric (0,2) tensor	RHS of Einstein field equations (GR); encodes matter/energy distribution
$T^{\text{ent}}_{\mu\nu}$	Entropic stress-energy tensor	Symmetric (0,2) tensor	Second moment of entropic momentum distribution; ToE generalization of $T_{\mu\nu}$
$R^\rho_{\sigma\mu\nu}$	Riemann curvature tensor	Type (1,3) tensor	Measures path-dependence of parallel transport; full curvature of M
$R_{\mu\nu}$	Ricci tensor	Symmetric (0,2) tensor	$R^\rho_{\mu\rho\nu}$; average curvature; LHS of EFE (partial)
R	Ricci scalar	Scalar field	$g^{\mu\nu}R_{\mu\nu}$; total scalar curvature; integrand of EH action
$G_{\mu\nu}$	Einstein tensor	Symmetric (0,2) tensor	$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$; divergence-free; LHS of EFE
C_{ijk}	Amari-Čencov cubic tensor	Symmetric (0,3) tensor on Λ	Third moment of score function; controls α -connection departure; measures irreversibility
$\Sigma^{\mu\nu}$	Covariance tensor	Symmetric (2,0) tensor	Second central moment of a distribution; positive semi-definite

16.5 Connections, Curvature, and Actions

Symbol	Name	Definition / ToE Role
$\Gamma_{\mu\nu}^{\rho}$	Christoffel symbols (Levi-Civita)	$\frac{1}{2}g^{\rho\sigma}(\partial_{\mu}g_{\nu\sigma} + \partial_{\nu}g_{\mu\sigma} - \partial_{\sigma}g_{\mu\nu})$; connection coefficients of Obidi Metric
∇_{μ}	Covariant derivative (Levi-Civita)	Coordinate-invariant derivative; $\nabla_{\mu}V^{\nu} = \partial_{\mu}V^{\nu} + \Gamma_{\mu\rho}^{\nu}V^{\rho}$
$\nabla^{(\alpha)}$	Amari-Čencov α -connection	Family of connections on Λ indexed by $\alpha \in \mathbb{R}$; $\alpha=0$ is Levi-Civita
$\text{OCI} = \ln 2$	Obidi Curvature Invariant	Minimum distinguishability curvature of the entropic manifold; quantum of informational curvature
A_{Obidi}	Obidi Action	Fundamental action functional of ToE; source of MEE and $T_{\mu\nu}^{\text{ent}}$
MEE	Master Entropic Equation	Field equation for $S(x)$ derived by $\delta A_{\text{Obidi}}/\delta S = 0$
HMAS	Hybrid Metric-Affine Space	Bundle structure combining classical (Fisher) + quantum (Fubini-Study) sectors

16.6 Probability and Information Theory

Symbol	Name	Definition / ToE Role
$p(x \theta)$	Parametric probability distribution	Distribution labeled by parameter $\theta \in \Lambda$; defines the statistical manifold
$\theta = (\theta^A)$	Statistical parameters	Coordinates on the statistical manifold Λ
$H(p)$	Shannon / differential entropy	$-\int p \log p \, dx$; the entropy field value $S(x) = H(p_x)$
$S(x)$	Entropy field	Smooth scalar field on M ; primary dynamical variable of ToE; $S(x) = H(p_x)$
$I(\theta)$	Fisher information	$\langle (\partial_\theta \log p)^2 \rangle$; scalar version of G^F_{AB} in 1D
$A(\theta)$	Log-partition function	$\ln \int \exp(\theta^i T_i(x)) dx$; $G^F_{ij} = \partial_i \partial_j A$ for exponential families
$f^{\text{ent}}(x, q)$	Entropic distribution function	Distribution over entropic momenta $q_\mu = \partial_\mu S$; derived from Obidi Action
$d\mu_F$	Fisher-Rao volume measure	$\sqrt{ G^F } \, d^n \theta$; invariant measure for fiber integrals on Λ

16.7 Physical Constants

Symbol	Name	Value / ToE Interpretation
G	Newton's gravitational constant	$6.674 \times 10^{-11} \, \text{m}^3 \text{kg}^{-1} \text{s}^{-2}$; emerges in ToE from OCI and Vuli-Ndlela normalization
c	Speed of light	$2.998 \times 10^8 \, \text{m/s}$; Entropic Speed Limit

Symbol	Name	Value / ToE Interpretation
		in ToE
$\hbar$	Reduced Planck constant	1.055×10^{-34} J·s; quantum of action; related to OCI = ln 2 in ToE
k_B	Boltzmann constant	1.381×10^{-23} J/K; conversion between information entropy and thermal energy
ℓ_P	Planck length	$(\hbar G/c^3)^{1/2} \approx 1.616 \times 10^{-35}$ m; scale of quantum gravitational effects
Λ_{cc}	Cosmological constant	Approximately 1.089×10^{-52} m ⁻² ; vacuum entropy density in ToE



References for Part I

The following references are the primary sources underlying the results surveyed in Part I. All references to ToE Living Review Letters I through III refer to the canonical archive.

A. Foundational Papers: Information Theory and Thermodynamics

[1] Shannon, C. E. (1948). A Mathematical Theory of Communication. *Bell System Technical Journal*, 27(3), 379-423; 27(4), 623-656. — Establishes the unique measure of information entropy (3.3) and the structural identity between information entropy and Boltzmann-Gibbs entropy.

[2] Landauer, R. (1961). Irreversibility and Heat Generation in the Computing Process. *IBM Journal of Research and Development*, 5(3), 183–191. — Establishes that erasing one bit of information dissipates at least $k_B T \ln 2$ of heat; grounds the information-primacy argument of Section 3.2(a).

[3] Wheeler, J. A. (1990). Information, Physics, Quantum: The Search for Links. In W. Zurek (Ed.), *Complexity, Entropy, and the Physics of Information*. Redwood City, CA: Addison-Wesley. — Original statement of the "It from Bit" conjecture; motivates the ToE revolutionary inversion.

B. Thermodynamic Gravity

[4] Bekenstein, J. D. (1973). Black Holes and Entropy. *Physical Review D*, 7(8), 2333–2346. — Establishes $S_{BH} = k_B A / (4\ell_P^2)$; fundamental to the thermodynamic gravity argument Section 3.2(b).

[5] Hawking, S. W. (1975). Particle Creation by Black Holes. *Communications in Mathematical Physics*, 43(3), 199–220. — Derives the Hawking temperature $T_H = \hbar c^3 / (8\pi G M k_B)$; establishes that gravitational geometry has thermodynamic properties.

[6] Jacobson, T. (1995). Thermodynamics of Spacetime: The Einstein Equation of State. *Physical Review Letters*, 75(7), 1260–1263. — Derives the full nonlinear Einstein field equations from the thermodynamic relation $\delta Q = T dS$ applied to local Rindler horizons. The most direct statement that GR is thermodynamics.

[7] Verlinde, E. (2011). On the Origin of Gravity and the Laws of Newton. *Journal of High Energy Physics*, 2011(4), 29. — Derives Newtonian gravity and Einstein equations as entropic forces from holographic screens; directly motivates ToE's entropic force mechanism.

C. Information Geometry

[8] Fisher, R. A. (1925). Theory of Statistical Estimation. *Mathematical Proceedings of the Cambridge Philosophical Society*, 22(5), 700–725. — Original definition of Fisher information; precursor to the Fisher-Rao metric.

- [9] Rao, C. R. (1945). Information and the Accuracy Attainable in the Estimation of Statistical Parameters. *Bulletin of the Calcutta Mathematical Society*, 37, 81–91. — Introduces the Fisher-Rao metric as the natural Riemannian metric on statistical manifolds; establishes the Cramér-Rao bound (14.11).
- [10] Čencov, N. N. (1982). *Statistical Decision Rules and Optimal Inference*. American Mathematical Society, Providence. (Translation of 1972 Russian original.) — Proves the uniqueness of the Fisher-Rao metric among metrics invariant under Markov congruent embeddings; the information-geometry argument of Section 3.2(c).
- [11] Amari, S. (1985). *Differential-Geometrical Methods in Statistics*. Lecture Notes in Statistics, Vol. 28. Springer-Verlag, Berlin. — Introduces the α -connections (Section 12.7), the dually flat geometry of exponential families, and the Amari-Čencov cubic tensor.
- [12] Amari, S., and Nagaoka, H. (2000). *Methods of Information Geometry*. American Mathematical Society / Oxford University Press. — The comprehensive reference for information geometry, covering Fisher-Rao metric, α -connections, duality, exponential families, and the geometry of quantum states.
- [13] Murray, M. K., and Rice, J. W. (1993). *Differential Geometry and Statistics*. Chapman and Hall, London. — Provides the differential geometric treatment of statistical manifolds, fiber bundles in statistics, and the geometric structure of estimation theory.

D. General Relativity and Mathematical Physics

- [14] Misner, C. W., Thorne, K. S., and Wheeler, J. A. (1973). *Gravitation*. W. H. Freeman and Company, San Francisco. — The comprehensive classical reference for general relativity; standard source for the Riemann tensor (Chapter 11), Einstein equations (Chapter 17), and Bianchi identities (Section 15.2).
- [15] Wald, R. M. (1984). *General Relativity*. University of Chicago Press. — The modern rigorous treatment of GR; primary reference for the differential geometry of Section 5 through 13, the Levi-Civita connection (Section 3.1), and the contracted Bianchi identity (Appendix B).

[16] Kobayashi, S., and Nomizu, K. (1963). *Foundations of Differential Geometry*, Vol. I. Wiley Interscience, New York. — The comprehensive reference for fiber bundles (Section 9), connections on principal bundles, and the curvature of connections.

[17] Lovelock, D. (1971). The Einstein Tensor and Its Generalizations. *Journal of Mathematical Physics*, 12(3), 498-501. — Proves uniqueness of the Einstein tensor as the divergence-free, symmetric, second-rank tensor expressible in terms of metric and its derivatives in 4 dimensions.

[18] Nash, J. (1956). The Imbedding Problem for Riemannian Manifolds. *Annals of Mathematics*, 63(1), 20-63. — Nash embedding theorem; underlies the induced metric equation (11.12) by which the Fisher-Rao geometry induces the Obidi Metric via the emergence map.

E. ToE Living Review Letters Series (Canonical Archive)

[19] Obidi, J. O. (2024/2025). ToE Living Review Letters I: The Theory of Entropicity — Foundations, the Obidi Action, and the Entropic Field. *Theory of Entropicity (ToE) Living Review Letters Series*. Letter I. Canonical Archive: entropicity.github.io/Theory-of-Entropicity-ToE/. — Original statement of the revolutionary inversion, the Obidi Action, and the Master Entropic Equation.

[20] Obidi, J. O. (2025). ToE Living Review Letters II: The Obidi Transformation, the Obidi Metric, and the Emergence of the Lorentzian Spacetime Geometry. *Theory of Entropicity (ToE) Living Review Letters Series*. Letter II. Canonical Archive: entropicity.github.io/Theory-of-Entropicity-ToE/. — Derives the Obidi Transformation converting the Fisher-Rao metric to the Obidi (Lorentzian) Metric; introduces the Entropic Speed Limit c .

[21] Obidi, J. O. (2025). ToE Living Review Letters IIA-C: The Entropic Speed Limit, the Hybrid Metric-Affine Space, and the Vuli-Ndlela Integral. *Theory of Entropicity (ToE) Living Review Letters Series*. Letter IIA-C. Canonical Archive: entropicity.github.io/Theory-of-Entropicity-ToE/. — Establishes c as the Entropic Speed

Limit; defines the HMAS with classical Fisher and quantum Fubini-Study sectors; introduces the Vuli-Ndlela Integral.

[22] Obidi, J. O. (2025). ToE Living Review Letters III: The Obidi Curvature Invariant, Mass Generation, and the Right-Hand Side of the Einstein Field Equations. *Theory of Entropicity (ToE) Living Review Letters Series*. Letter III. Canonical Archive: entropicity.github.io/Theory-of-Entropicity-ToE/. — Establishes $\text{OCI} = \ln 2$ as the quantum of informational curvature; derives mass from entropic localization; derives the Entropic Stress-Energy Tensor.

F. Additional References

[23] Kullback, S., and Leibler, R. A. (1951). On Information and Sufficiency. *Annals of Mathematical Statistics*, 22(1), 79–86. — Introduces the Kullback-Leibler divergence; used in Parts II and III for the divergence geometry of ToE.

[24] Araki, H. (1976). Relative Entropy of States of Von Neumann Algebras. *Publications of the Research Institute for Mathematical Sciences*, 11(3), 809–833. — Defines the Araki-Umegaki quantum relative entropy; relevant for the quantum sector of the HMAS.

[25] Rényi, A. (1961). On Measures of Entropy and Information. In *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*, Vol. 1, pp. 547–561. — Introduces the Rényi entropy family; used in ToE for the characterization of non-extensive entropic systems.

[26] Tsallis, C. (1988). Possible Generalization of Boltzmann-Gibbs Statistics. *Journal of Statistical Physics*, 52(1-2), 479–487. — Introduces Tsallis (non-extensive) entropy; complementary to Rényi entropy; relevant for far-from-equilibrium entropic systems in ToE.

[27] de Groot, S. R., and Mazur, P. (1962). *Non-Equilibrium Thermodynamics*. North-Holland, Amsterdam. — The classical reference for the local entropy density as a dynamical field (3.8); direct precursor to ToE's promotion of $S(x)$ to a primary field.

[28] Cercignani, C., and Medeiros Kremer, G. (2002). *The Relativistic Boltzmann Equation: Theory and Applications*. Birkhäuser, Basel. — Reference for the relativistic kinetic theory fiber integral formula (14.8) for the stress-energy tensor.



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<https://entropicity.github.io/Theory-of-Entropicity-ToE/>

*Part II: Information Geometry and
ToE Core Structure Part III:
Derivation of the Einstein Field
Equations and Comprehensive
Literature Review — forthcoming*

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End of Part I — ToE Living Review Letters IV — Theory of Entropicity — June 2026

Note on Notation: All equations involving the Einstein summation convention are written with the convention that repeated indices, one raised and one lowered, are summed over their full range. Spacetime indices (μ, ν, ρ, σ) run from 0 to 3. Statistical manifold indices (A, B, C) run from 1 to n where n is the dimension of the statistical manifold. The metric signature convention is $(-, +, +, +)$ throughout. Natural units with $c = 1$ and $k_B = 1$ are used in intermediate computations unless physical clarity requires restoration of constants. All ToE-specific quantities (Obidi Action, Obidi Metric, Obidi Transformation, Obidi Curvature Invariant, Master Entropic Equation, Vulindlela Integral, Hybrid Metric-Affine Space, Entropic Stress-Energy Tensor) are original contributions of the Theory of Entropicity; they are defined precisely as they appear in this Letter and in the prior Letters of the ToE Living Review Letters Series, cited as [19]-[22] above.